

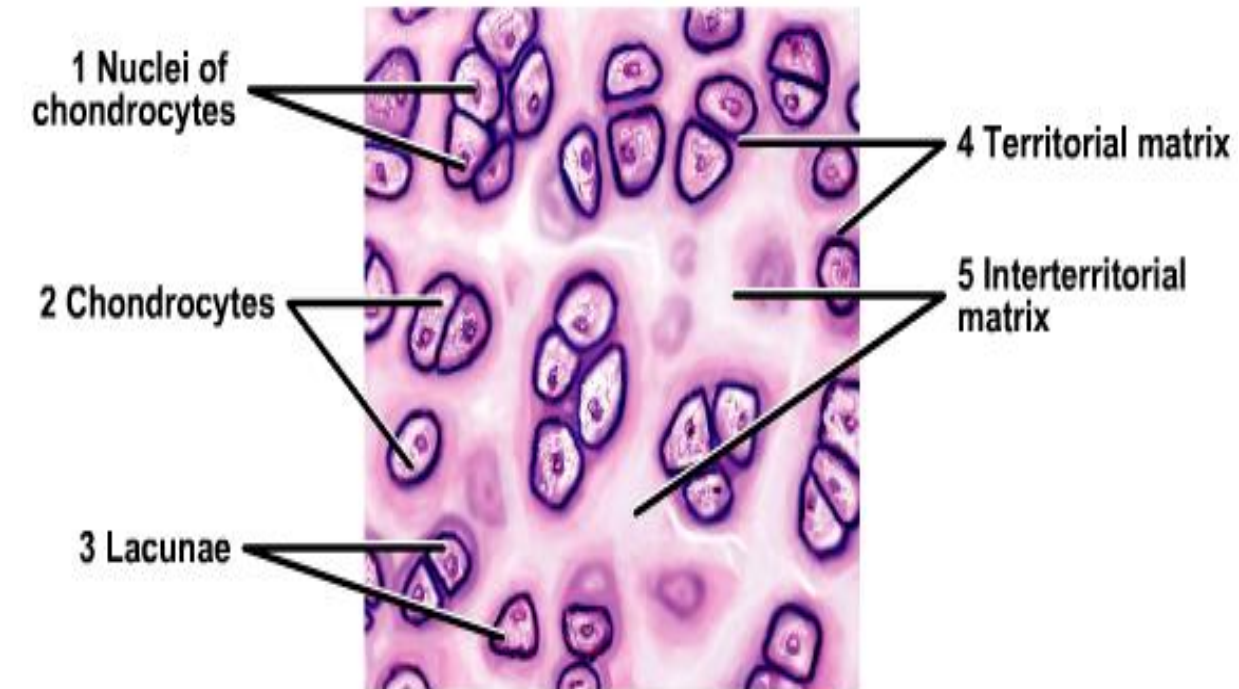
SKELETAL CONNECTIVE TISSUES

1. BONE TISSUE

2. CARTILAGE

3. LIGAMENTS

4. JOINTS



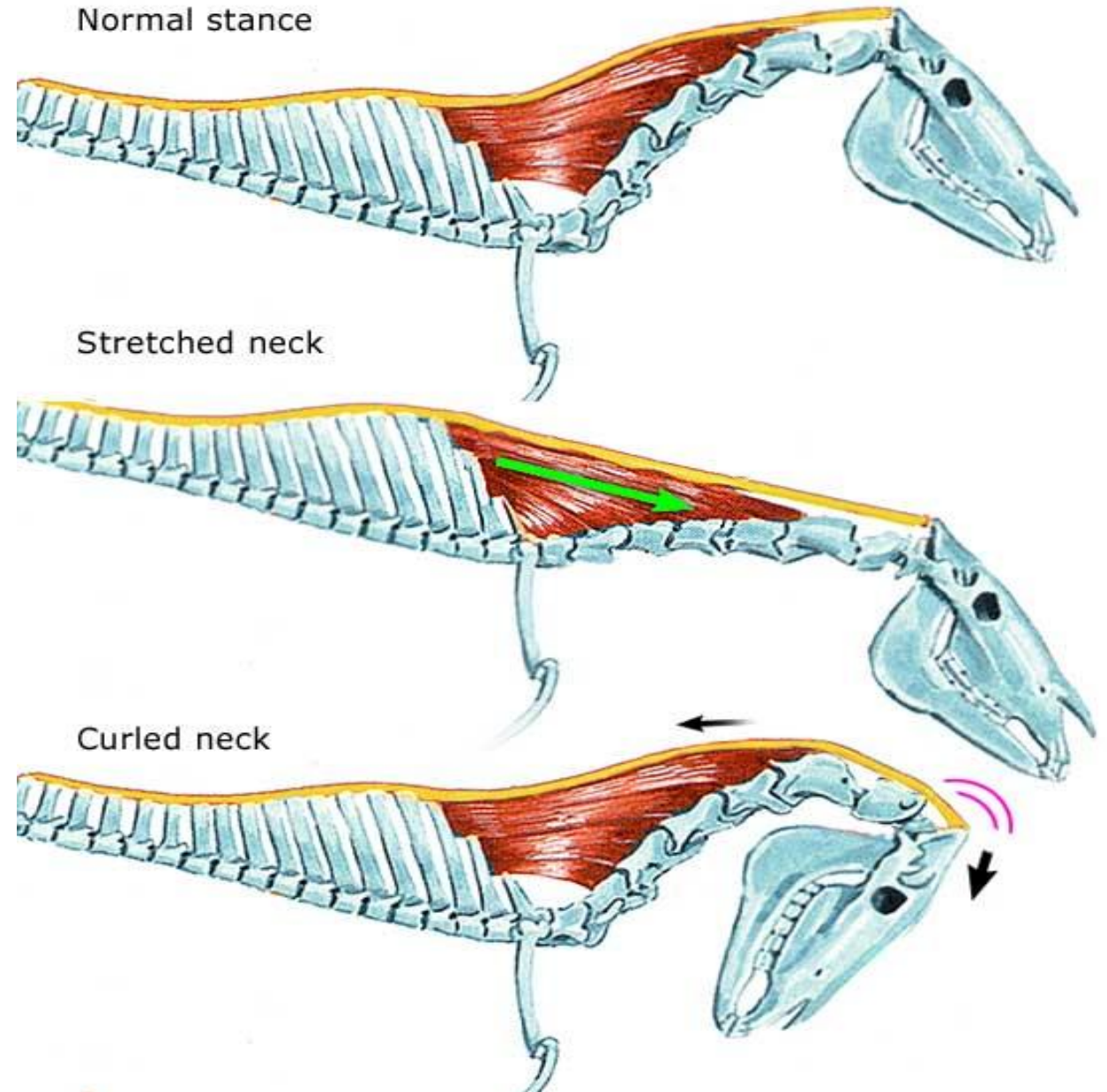
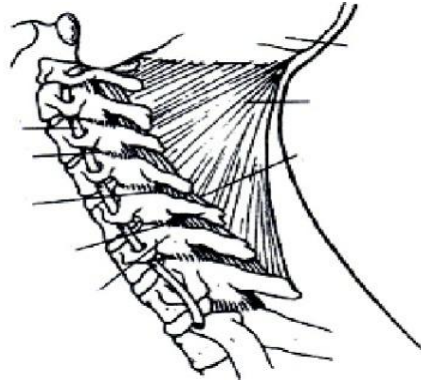
Dense Regular Collagenous Connective Tissue

- The collagen bundles are arranged according to a definite pattern, with the linear orientation of fibroblasts in response to prolonged stresses exerted in the same direction.
- This arrangement offers great resistance to traction forces.
- Examples such tissues include **Tendons** and **white ligaments**

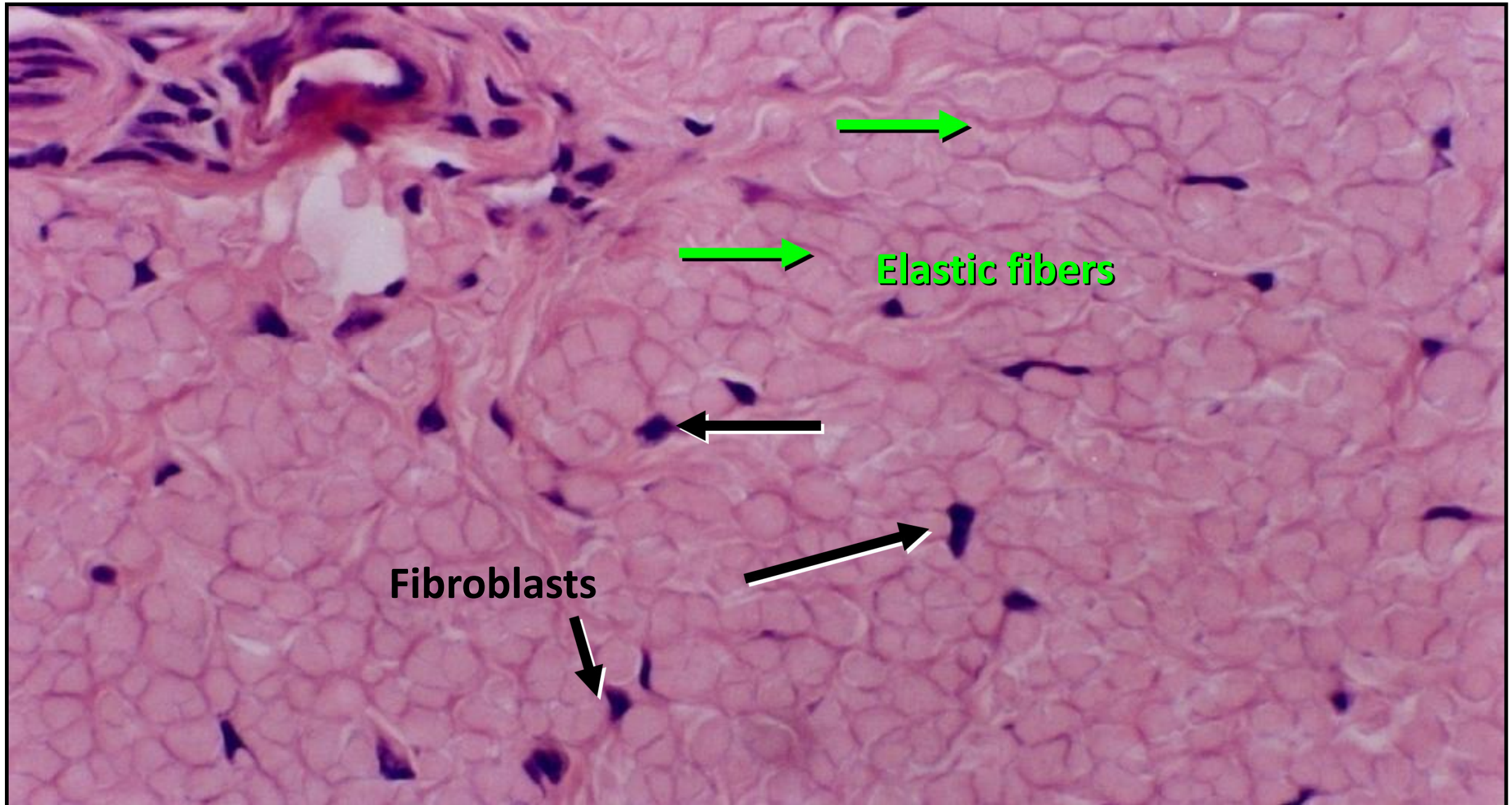


NOTE the waviness of the fibers. What function could this structural aspect provide?

Ligamentum nuchae (Elastic ligament)



Ligamentum nuchae (cross section)



CARTILAGES

- belongs to the skeletal (supportive) tissues which provide flexible support to some organs.

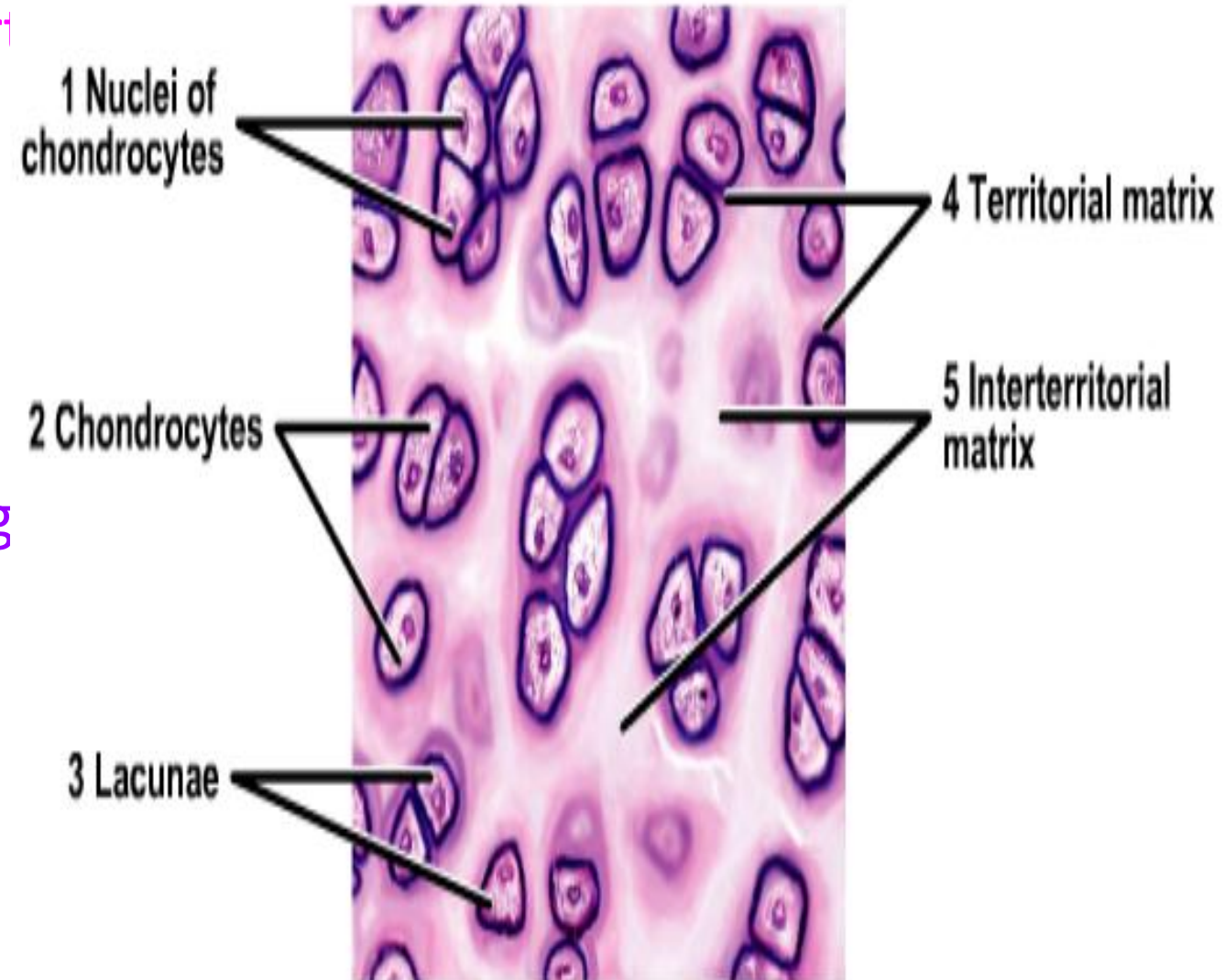
Cartilage consists of:-

1. Cartilage cells: chondroblasts and chondrocytes

- **chondrocytes** are located in **lacunae** surrounded by an intercellular cartilag matrix.

2. Extracellular matrix:

- Collagenous and elastic fibers**
- Ground substance: enriched with glycosaminoglycans and proteoglycan aggregates.**



- Cartilage is strong and weight bearing tissue with some degree of flexibility.
- Consists of **cells** and **extracellular matrix**.
- The matrix is a firm pliable matrix that resists mechanical stresses.
- The cartilage matrix is nonvascular and does not have nerve supply or lymph vessels.
 - nerves are not present in cartilage, but nerves and nerve ending are present in the perichondrium.
 - receives its nutrients from blood vessels in the surrounding dense connective tissue, the **perichondrium** or from synovial fluid.
 - cartilage is a tissue of very **low metabolic activity** and **slow cell turnover rate** (except in the embryo).
- It is involved in supporting the body because they are intimately associated in the skeletal system.
- Derived from mesenchyme (both ecto- and mesomesenchyme).

- The proteoglycans aggregates represent a complex of protein and sulfated and nonsulfated glycosaminoglycans (GAGs), particularly:
 - Hyaluronic acid (non-sulfated GAG)
 - Chondroitin-4-sulfate
 - Chondroitin-6-sulfate and
 - Keratan sulfate Type II (in fibrocartilages)
- The cartilage matrix also consists of small amounts of several structural multiadhesive glycoproteins including chondronectin.
 - Like fibronectin in ordinary CT, **chondronectin** binds specifically to GAGs, collagen type II and integrins, mediating the adherence of chondrocytes to the ECM.
- These sulfated GAGs provide the basophilic staining characteristics of cartilages and staining variations within the matrix reflect differences in the molecular composition.

- The high content of solvation water bound to the negative charges of the GAGs acts as a shock absorber or biomechanical spring; this is of great functional importance, especially in articular cartilages.
- Immediately surrounding the group of chondrocytes (isogenous group) the ECM is richer in GAGs and poor in collagenous and elastic fibers.
- These areas comprise the **territorial matrix** and usually stain differently from the rest of the matrix.

FUNCTIONS OF CARTILAGE

Cartilage is important for:

- provision of flexible skeletal support to the embryo prior to the development of the bony skeleton.
 - forms a template for most of the developing bony skeleton.
- elongation of developing long bones (endochondral ossification).
- facilitates smooth movement at joints (by articular cartilages).
- flexible support in the ear auricle and eartubes, and in the larger tubes of the respiratory tract (nose, larynx, trachea & bronchi).
- Cartilages maintain the patency of these air passageways

Characteristics of Cartilage

1. Cartilage is avascular (has no blood supply) so, healing is poor or slow after injury.
2. It has no lymphatic vessels or nerves either.
3. **Chondroblasts** and **chondrocytes** synthesize and secrete the cartilage matrix.
4. **Chondrocytes** are located in matrix cavities called **lacunae**.
5. Chondrocytes derived by mitotic division of one parent cell exist as clusters of 4 to 8 cells known as **isogenous groups (cell nests)**.
6. Cytoplasm of chondrocytes is basophilic, rich in glycogen, lipids and phosphatase enzymes.
7. Cartilage is usually covered or surrounded by a capsule-like sheath of dense CT known as **Perichondrium**, formed of two layers: outer fibrous layer & inner chondrogenic layer.

Trachea H&E

interterritorial
matrix

isogenous
group

territorial
matrix



1st cycle

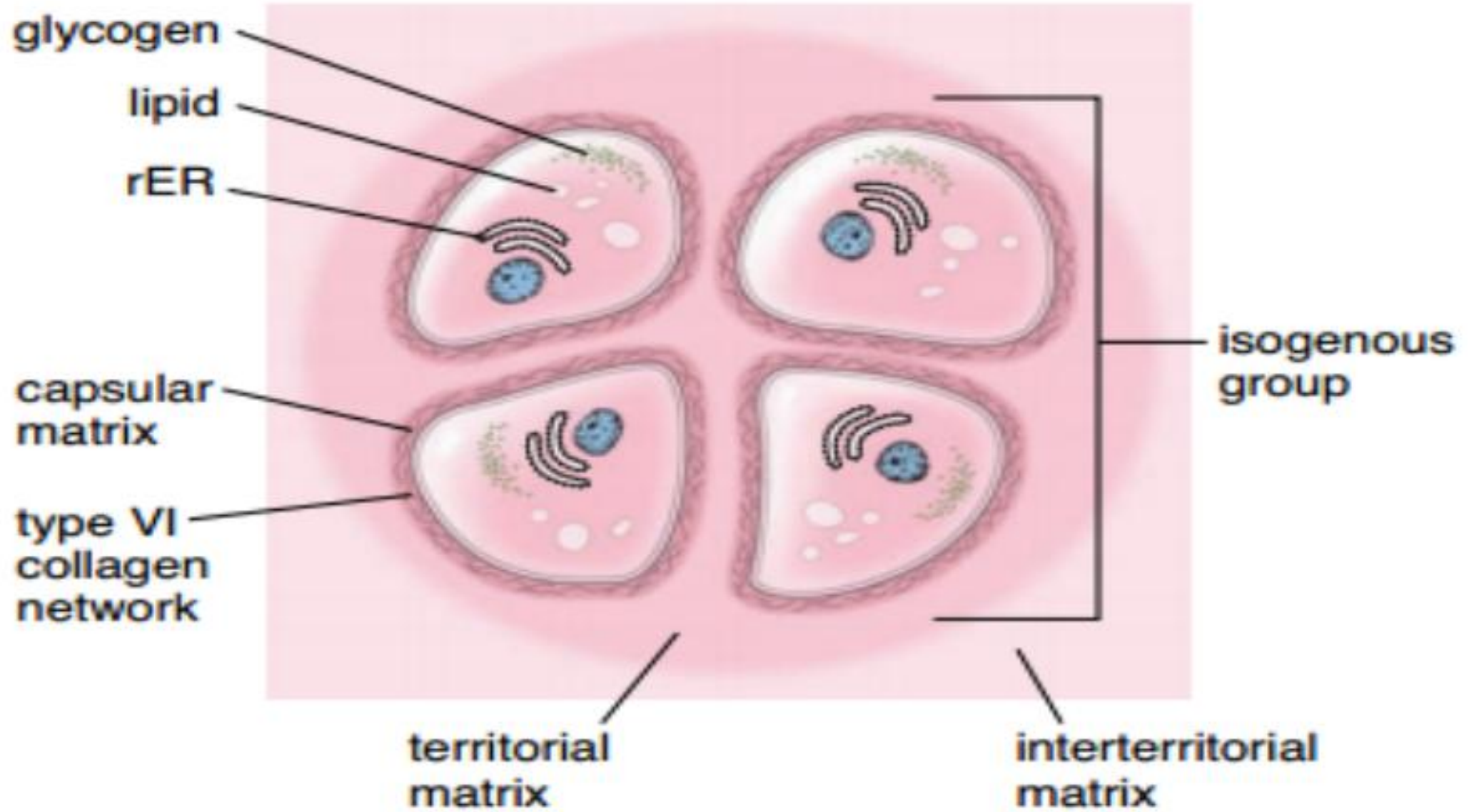


Diagram of cartilage matrices. Note the areas of capsular, territorial, and interterritorial matrices.

PERICHONDRIUM

- a capsule-like sheath of dense fibrous connective tissue surrounding the cartilage.
- houses blood vessels which supply cartilage cells by diffusion.
- *When actively growing, appears divided into an inner cellular layer and an outer fibrous layer.*

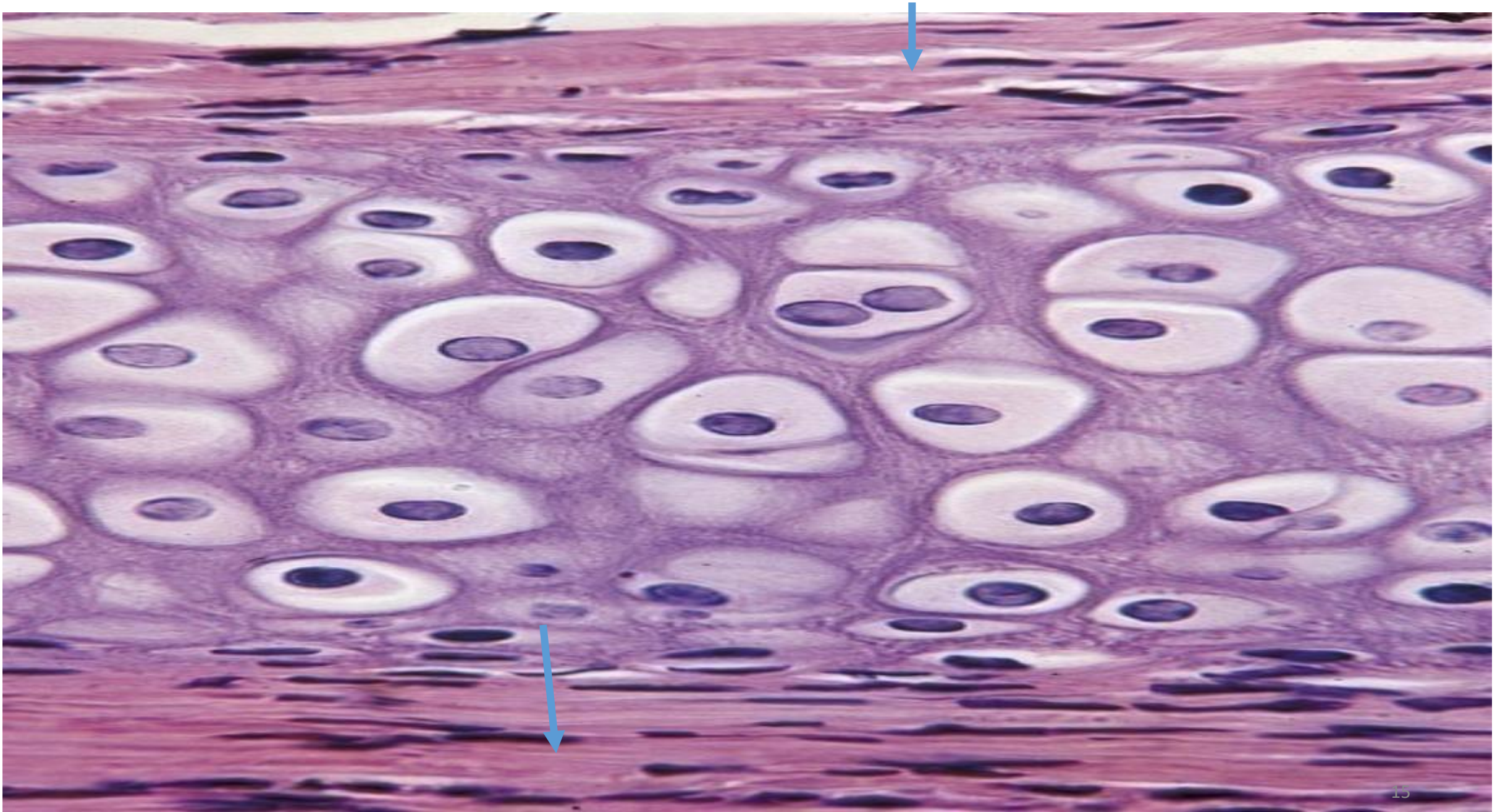
1. The outer fibrous layer:

- Is formed of fibrous CT containing *type I collagens and elastic fibers*; fibroblasts, fibrocytes & blood vessels.

2. The inner cellular layer (chondrogenic layer):

- is so called because it contains the cartilage forming cells, **chondroblasts**.
- important for appositional growth of the cartilage
- Avascular and consists of collagen type II fibrils

PERICHONDRIUM

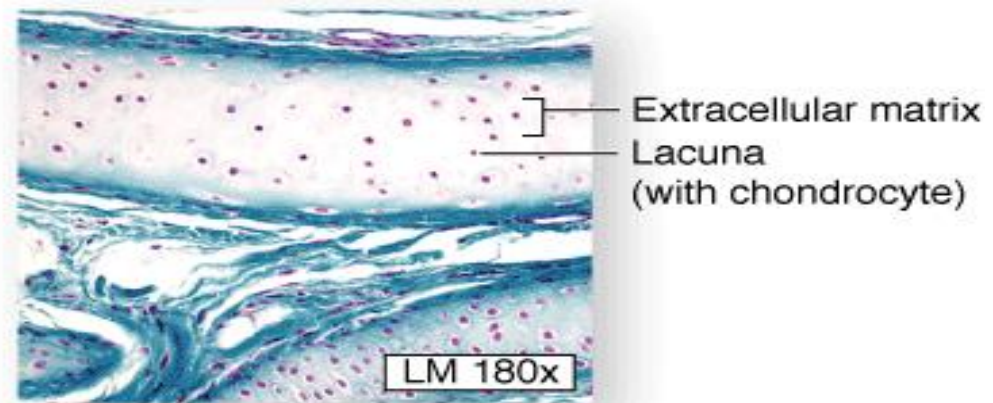
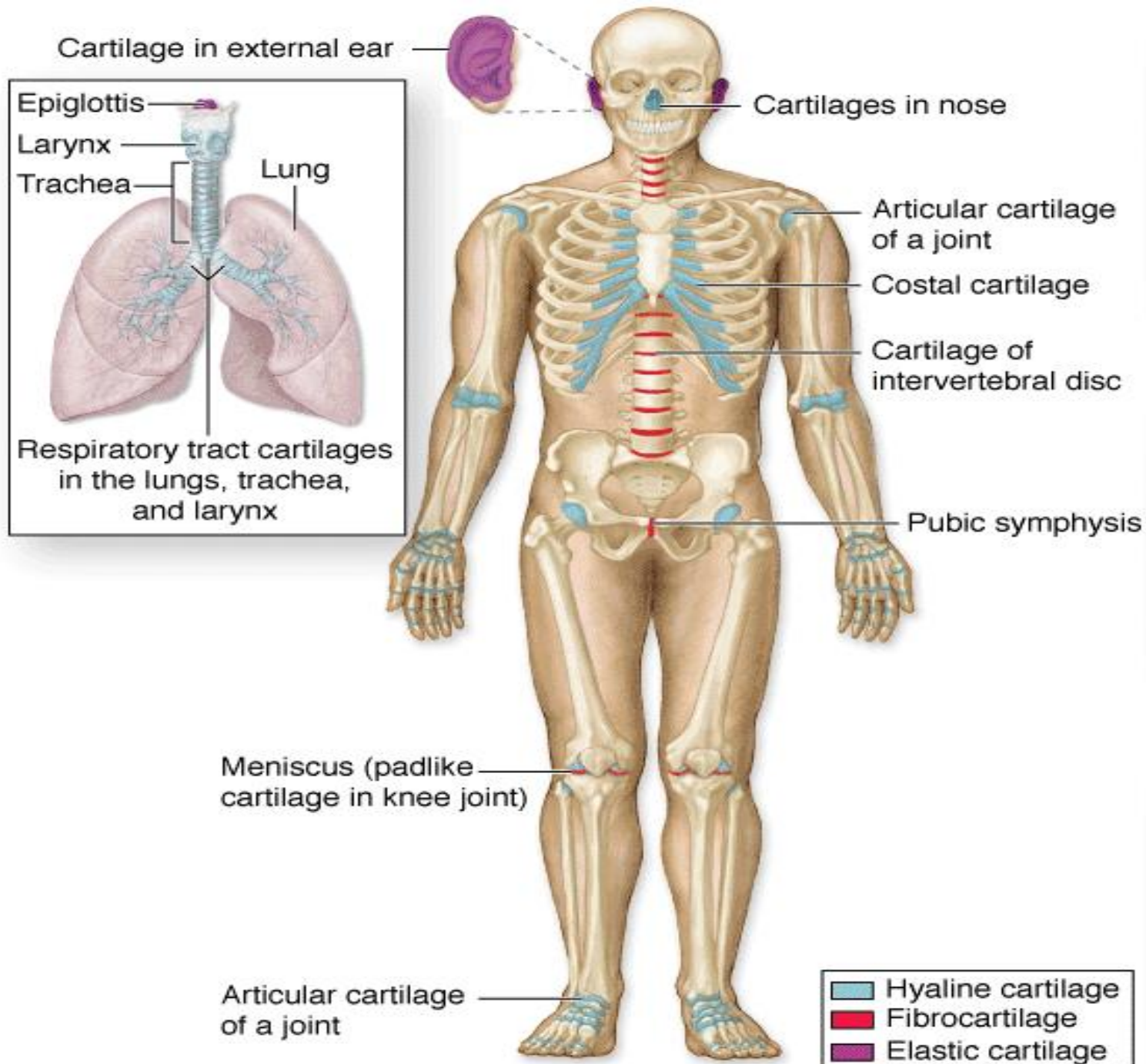


Hyaline cartilage, lacunae, chondrocytes, perichondrium, matrix

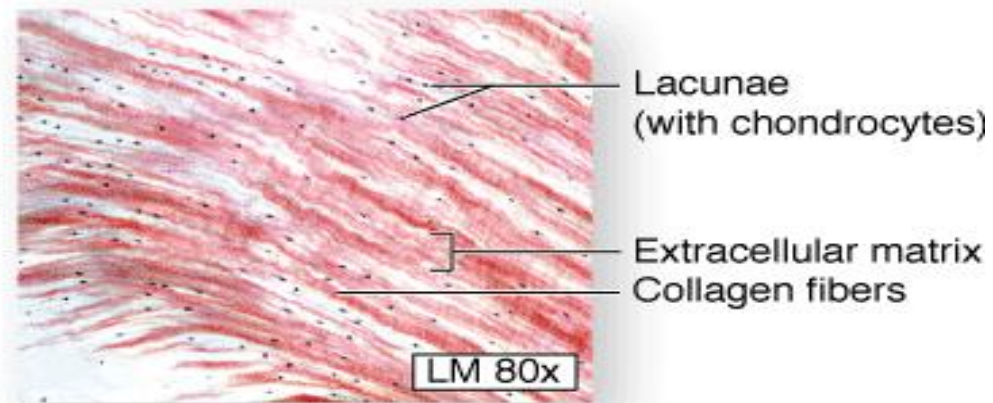


- Variations in the composition of cartilage matrix components produce three types of cartilages adapted to local biomechanical needs:

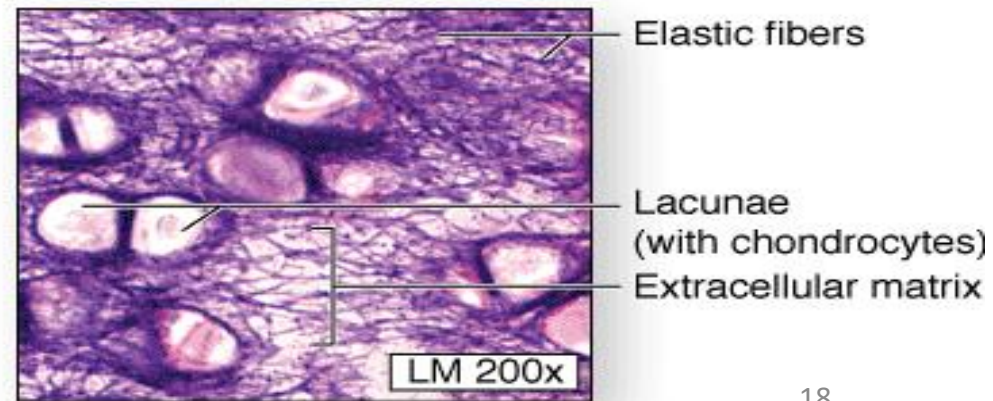
- 1. Hyaline cartilage**
- 2. Elastic cartilage**
- 3. Fibrocartilage**



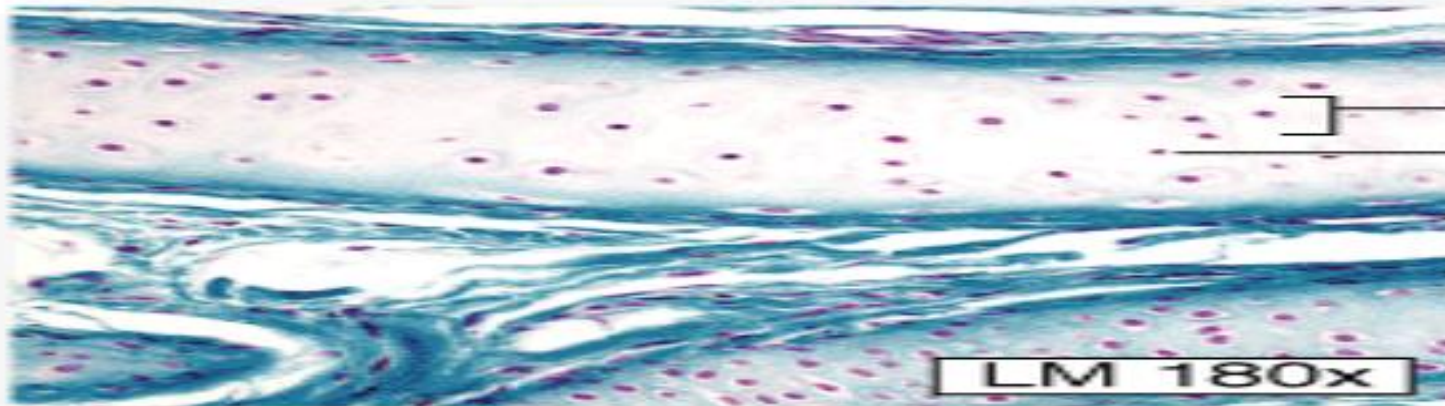
b Hyaline cartilage



c Fibrocartilage



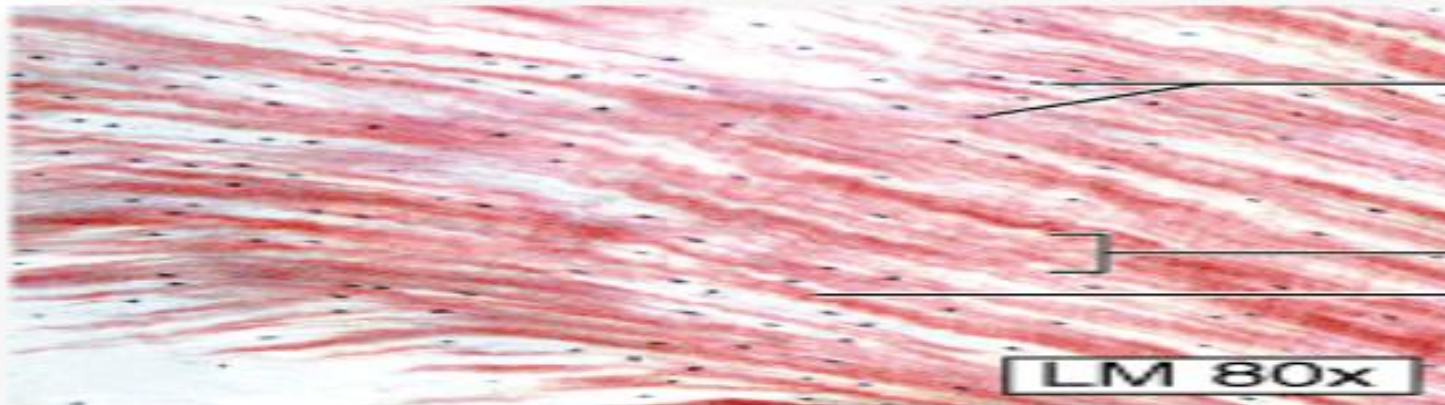
d Elastic cartilage



Extracellular matrix
Lacuna
(with chondrocyte)

[LM 180x]

b Hyaline cartilage

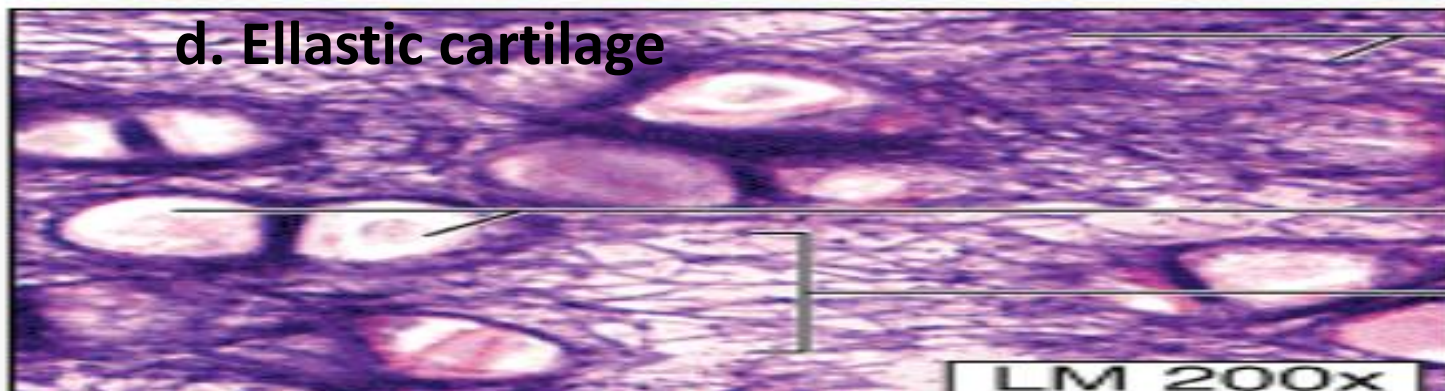


Lacunae
(with chondrocytes)

Extracellular matrix
Collagen fibers

[LM 80x]

c Fibrocartilage



d. Elastic cartilage

Elastic fibers

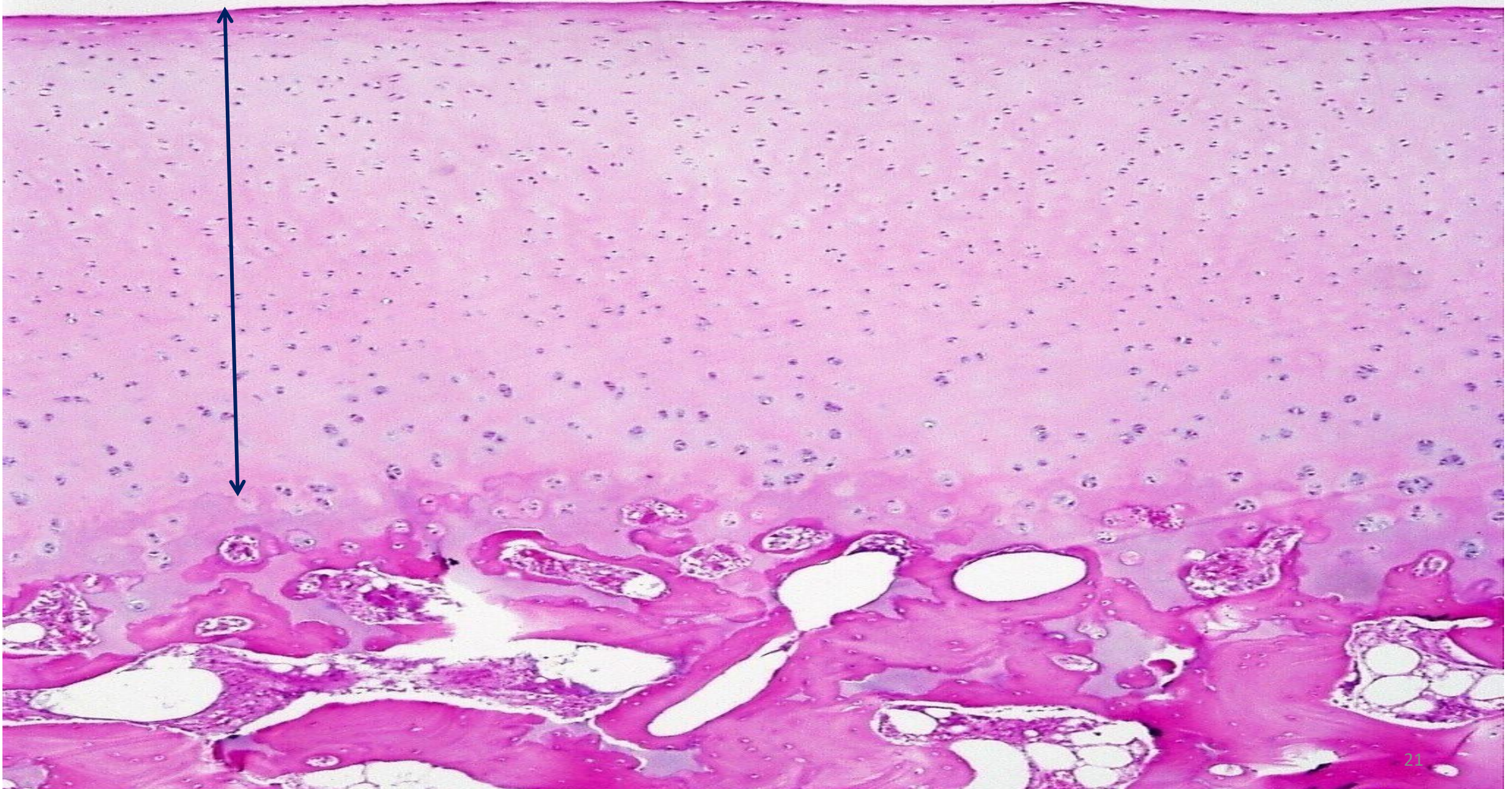
Lacunae
(with chondrocytes)
Extracellular matrix

[LM 200x]

HYALINE CARTILAGE

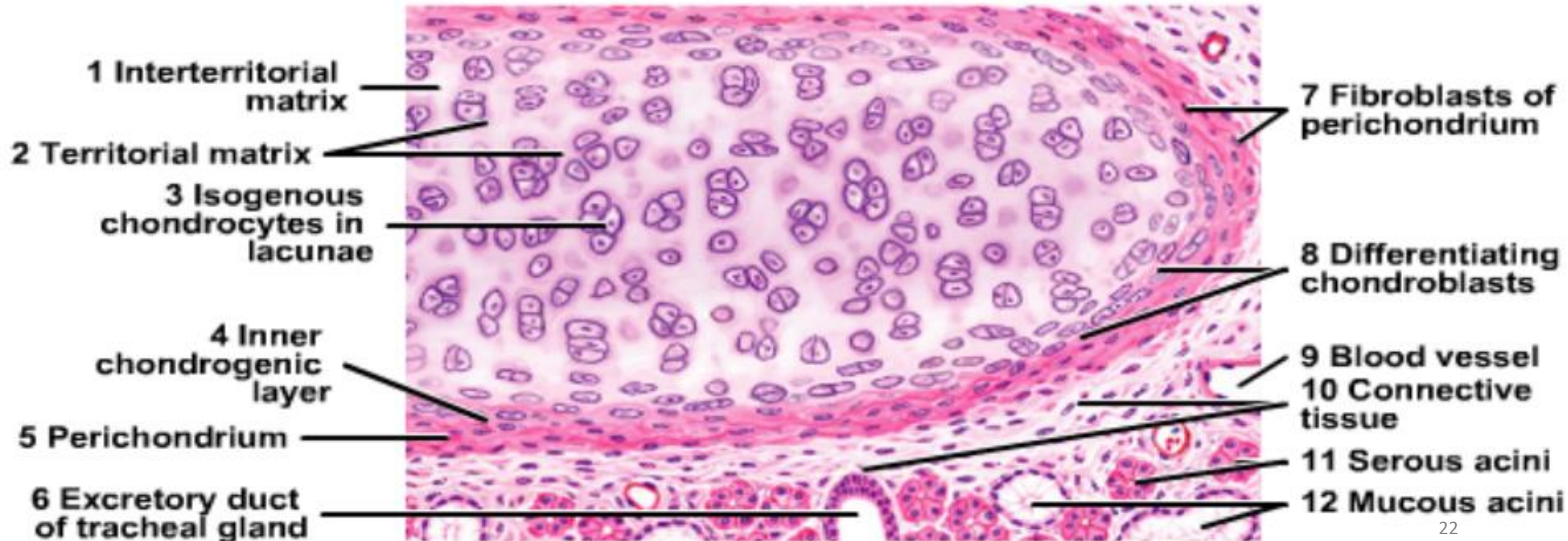
- is the most common form of cartilage, that is both flexible and resilient to mechanical forces.
- Fresh hyaline cartilage is a semi-transparent (translucent), homogenous, smooth & appears **milky** or **glassy-white**.
- In adults, hyaline cartilage is found in the respiratory tract (nose, larynx, trachea, bronchi), the sternal ends of ribs, and on articulating surfaces of long bones and joints (**articular cartilages**).
- is much more common in the embryo, where it plays an important role in long bone development.
- Approximately 60 to 80% of the wet weight of hyaline cartilage is water.
- Type II collagen fibers are embedded in the matrix and provide structural support & constitute about 40% of the dry weight of hyaline cartilage.

ARTICULAR CARTILAGE

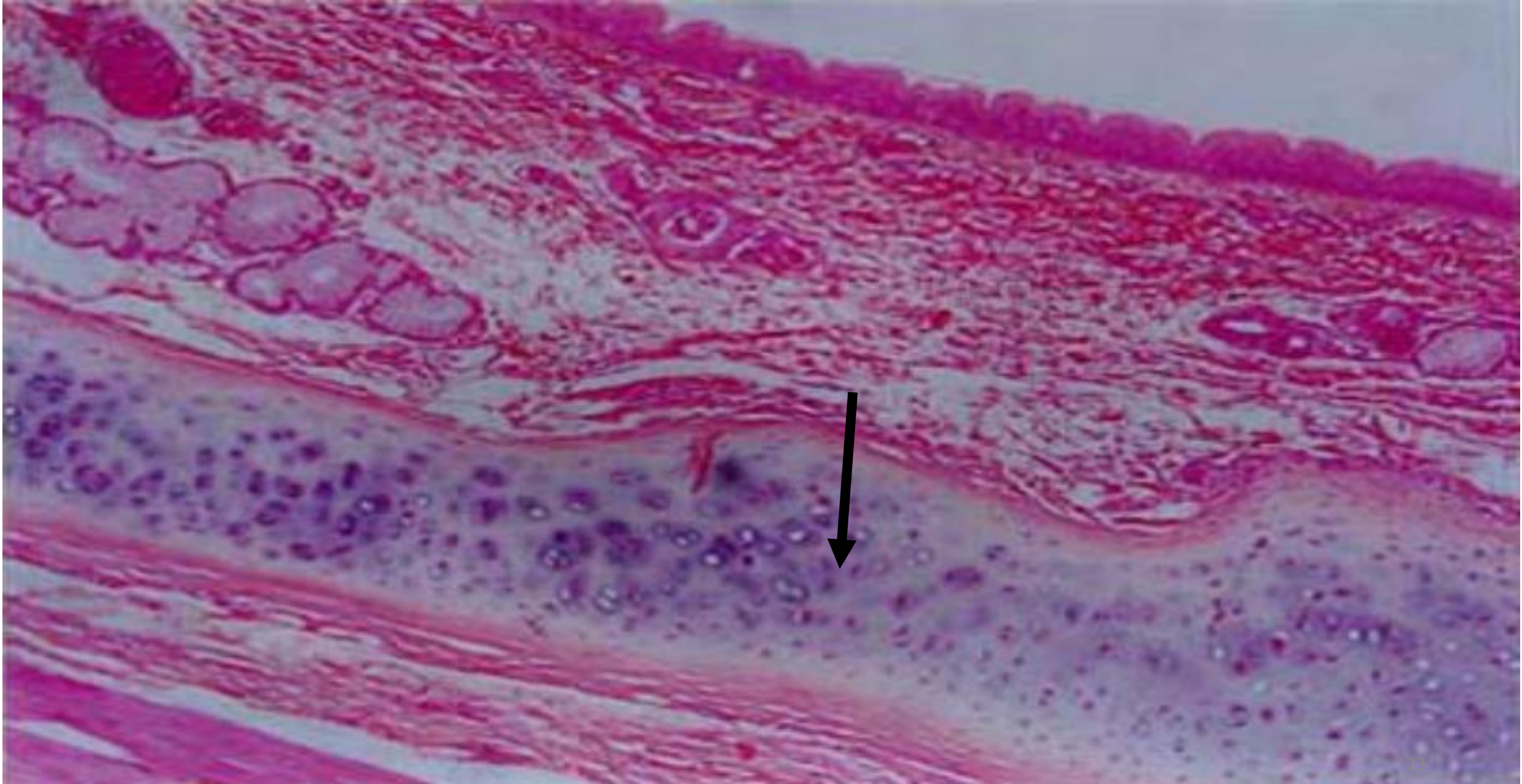


Locations sites of Hyaline Cartilages:

1. Costal cartilage
2. Trachea & bronchi
3. Skeleton of the fetus
4. Articular surface of bones
5. Nose
6. Larynx (Thyroid, cricoid & inferior parts of arytenoid cartilages)
7. Epiphyseal growth plates



TRACHEA, T.S. HYALINE CARTILAGE IN ITS WALL



ELASTIC CARTILAGE

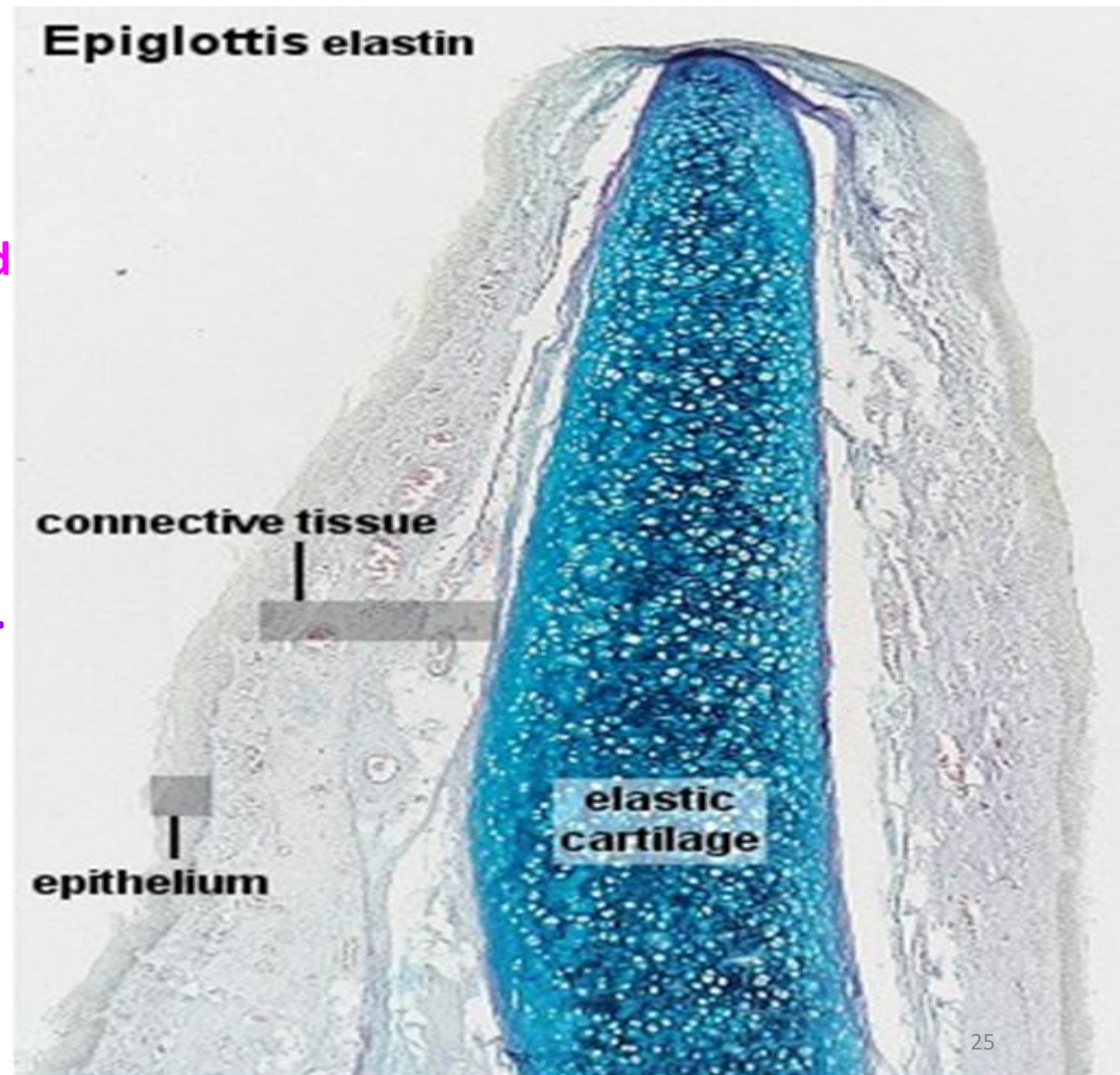
- is essentially very similar to hyaline cartilage except that it contains an abundant network of fine elastic fibers in addition to collagen type II fibrils.
- characterized by its great flexibility and elasticity owing to the large quantities of elastic fibers in the matrix.
- The elastin found in the elastic fibers impart a yellowish color to fresh elastic cartilage tissues.
- The elastic fibers in the matrix near the perichondrium are less-densely packed (and easier to see) than those deeper in the tissue.
- Elastic cartilage is found in the **external ear (pinna)**, in the **walls of the external auditory meatus** and **Eustachian tube** and also in the **epiglottis** and **cuneiform, corniculate**, and the **superior portions of arytenoid cartilages**).

Elastic Cartilage

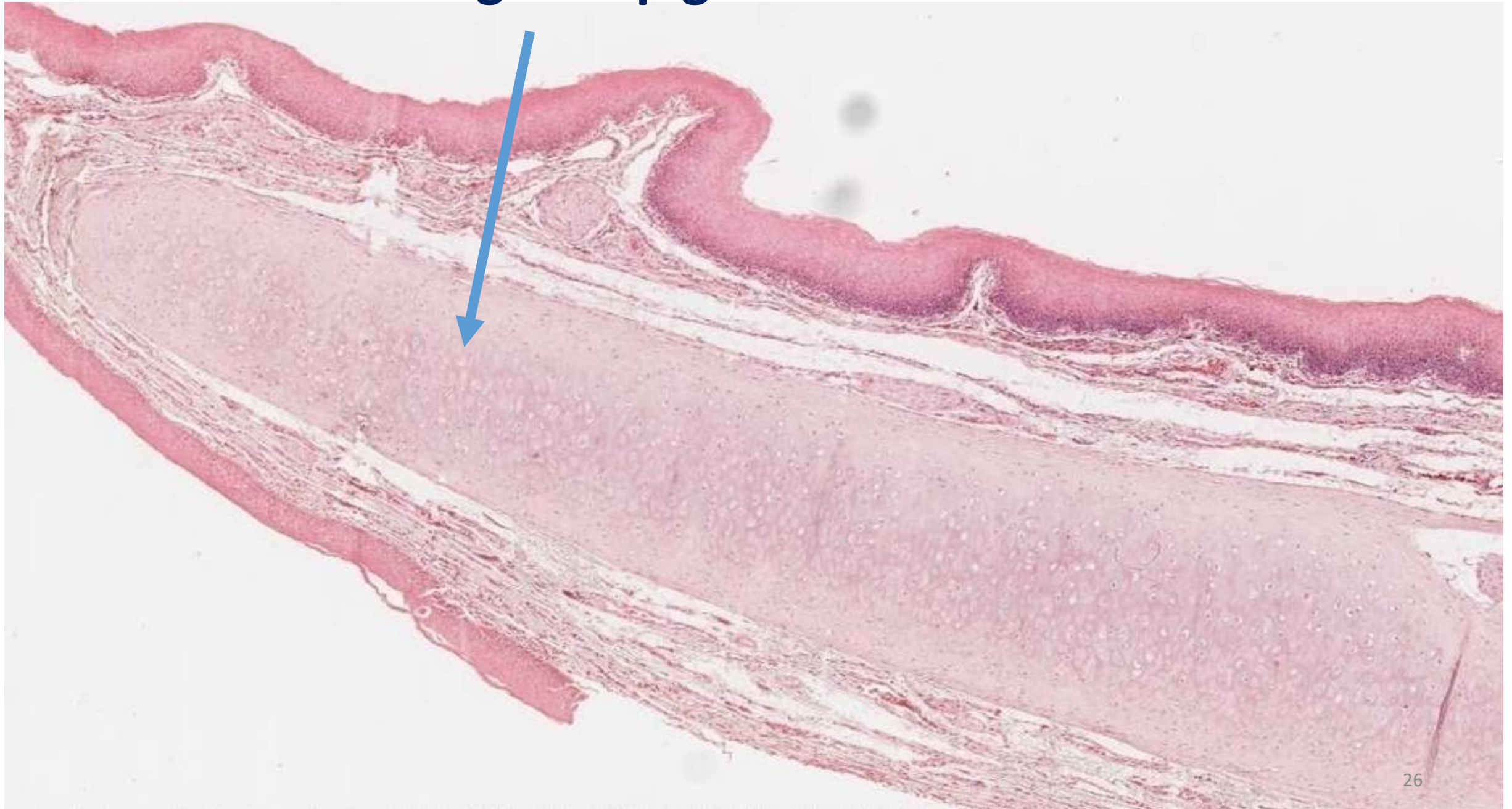
- Microscopic Appearance: similar to hyaline cartilages except presence fibrous elastic elements in its matrix :
- Elastic fibers form web-like mesh amid the lacunae.
- Always covered by a perichondrium.

Location and Distributions:

- External ear.
- Eustachian tube – connects the middle ear cavity to the nasopharynx.
- Larynx: Epiglottis, Cuneiform, Corniculate, and portions of Arytenoid cartilage
- **Epiglottis** – flap of organ that covers the inlet into the larynx during swallowing to prevent food/liquid from going down the “wrong pipe.”



Elastic Cartilage in Epiglottis



FIBROCARTILAGE

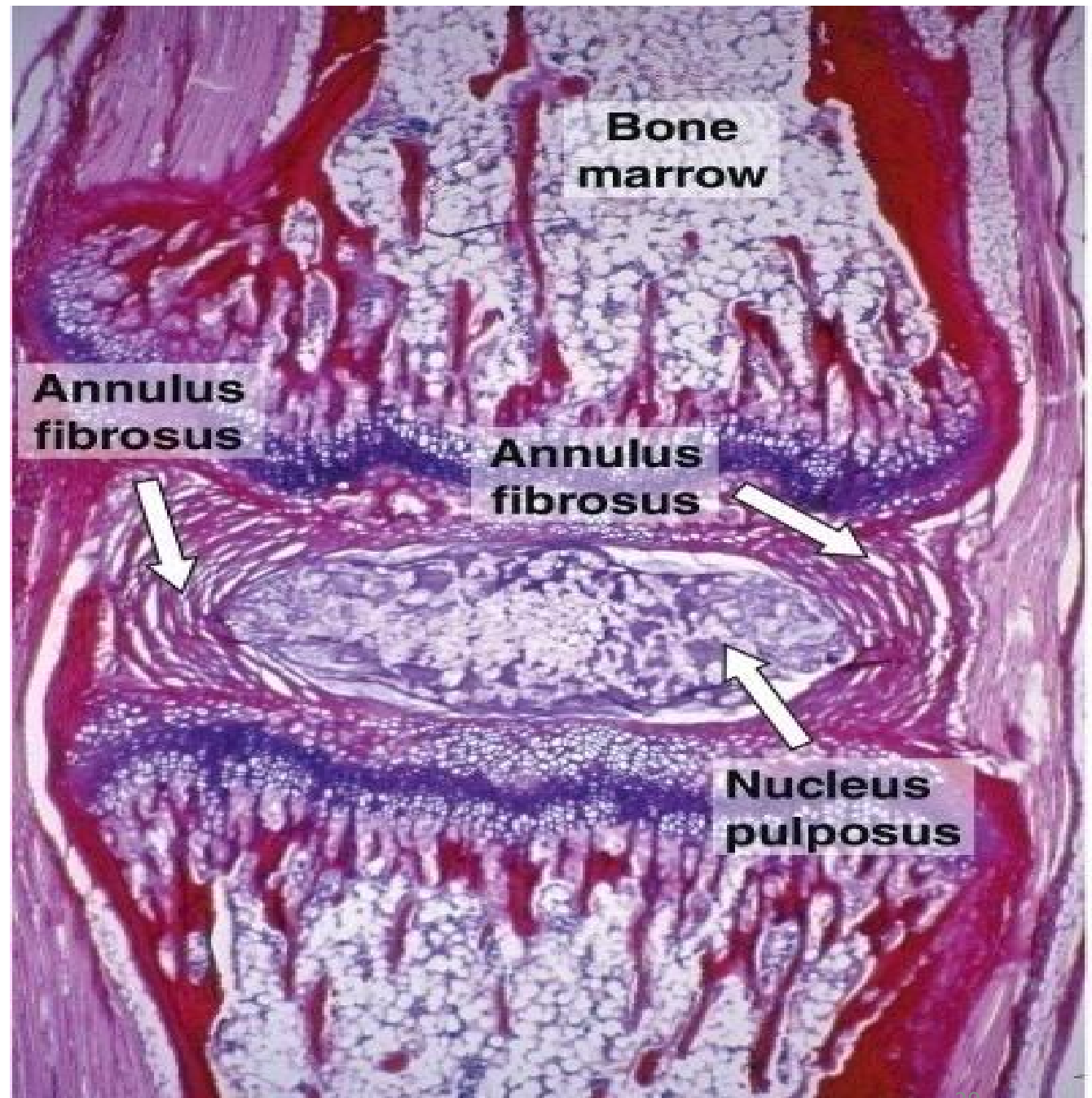
- transition between cartilage and dense CT in properties
- found in areas of the body subject to high mechanical stresses or weight bearing.
- It lacks the flexibility of the other cartilage types.
- Lacks perichondrium
- Fibrocartilages form the:
 - **Intervertebral discs**
 - **Pubic symphysis (Symphysis pubis)**
 - **Articular discs of some joints such as sternoclavicular joint, wrist joint and TMJ**
 - **Mensci of knee joints (the lateral & medial mensci)**
 - **Glenoid labrum and acetabular labrum** (fibro-cartilage pads which deepens the cavity of these joints)
 - attachments of certain ligaments and tendons

FIBROCARTILAGES (Arrow heads) pad between articulating bones at typical cartilaginous joint



An Intervertebral Disc (IVD):

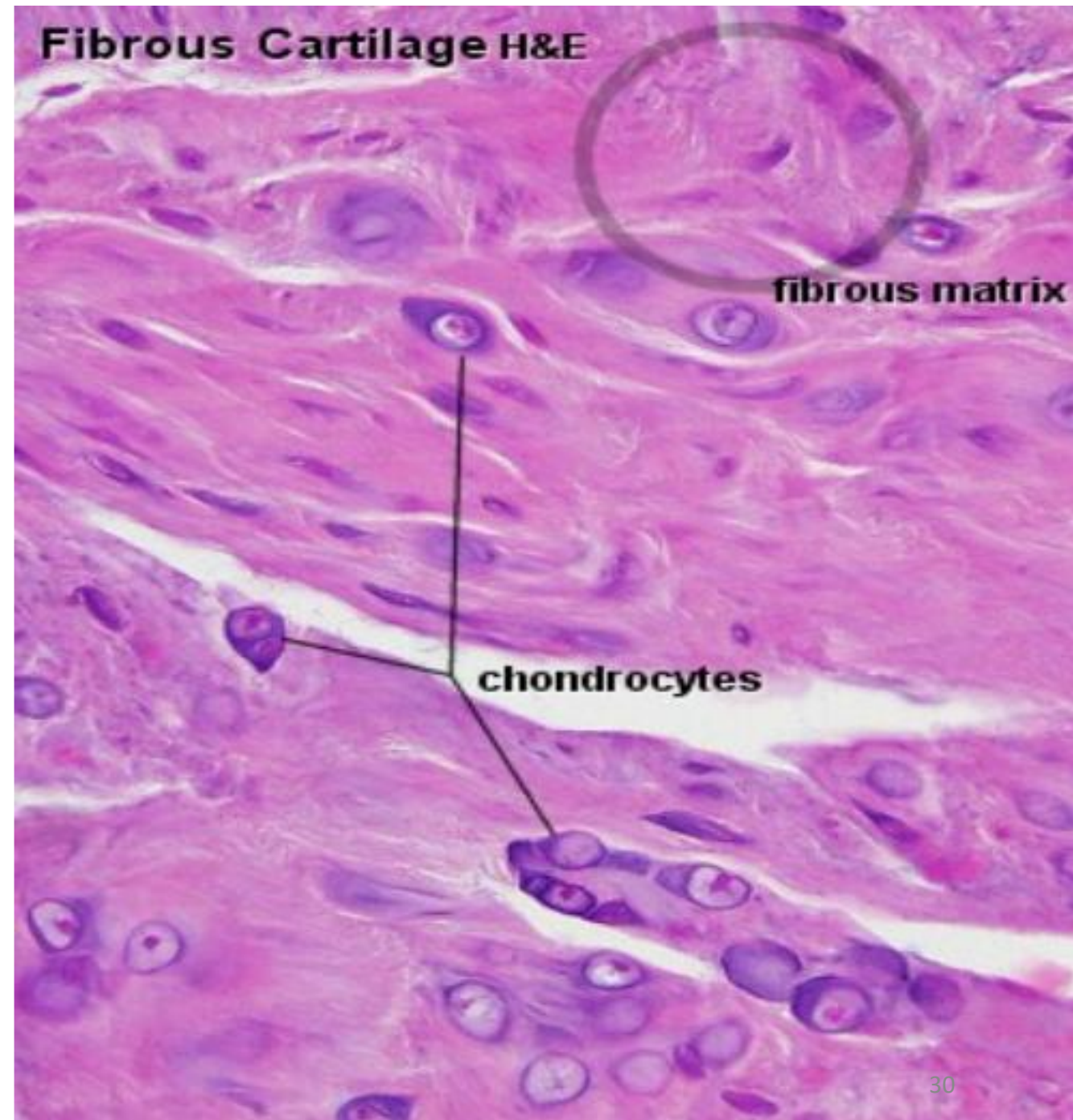
Section of a rat tail showing a small IVD and two adjacent vertebrae. The disc consists of concentric layers of fibrocartilage, comprising the **annulus fibrosus**, which surrounds the **nucleus pulposus**. The nucleus pulposus contains scattered residual/remnant cells of the embryonic notochord embedded in abundant gel-like matrix. In humans the IVDs have similar components but function primarily as shock absorbers within the spinal column, as well as making possible greater mobility of the spinal column.



Fibrocartilage

Microscopic Appearance:

- Parallel collagenous fibers (collagen type I) similar to those of dense CTs.
- Rows of chondrocytes in lacunae between collagenous fibers.
- **Chondrocytes** are fewer and smaller and are usually not in isogenous groups.
- lacks perichondrium



MEDICAL APPLICATION

1. Degenerative aging processes (Calcification) in cartilages

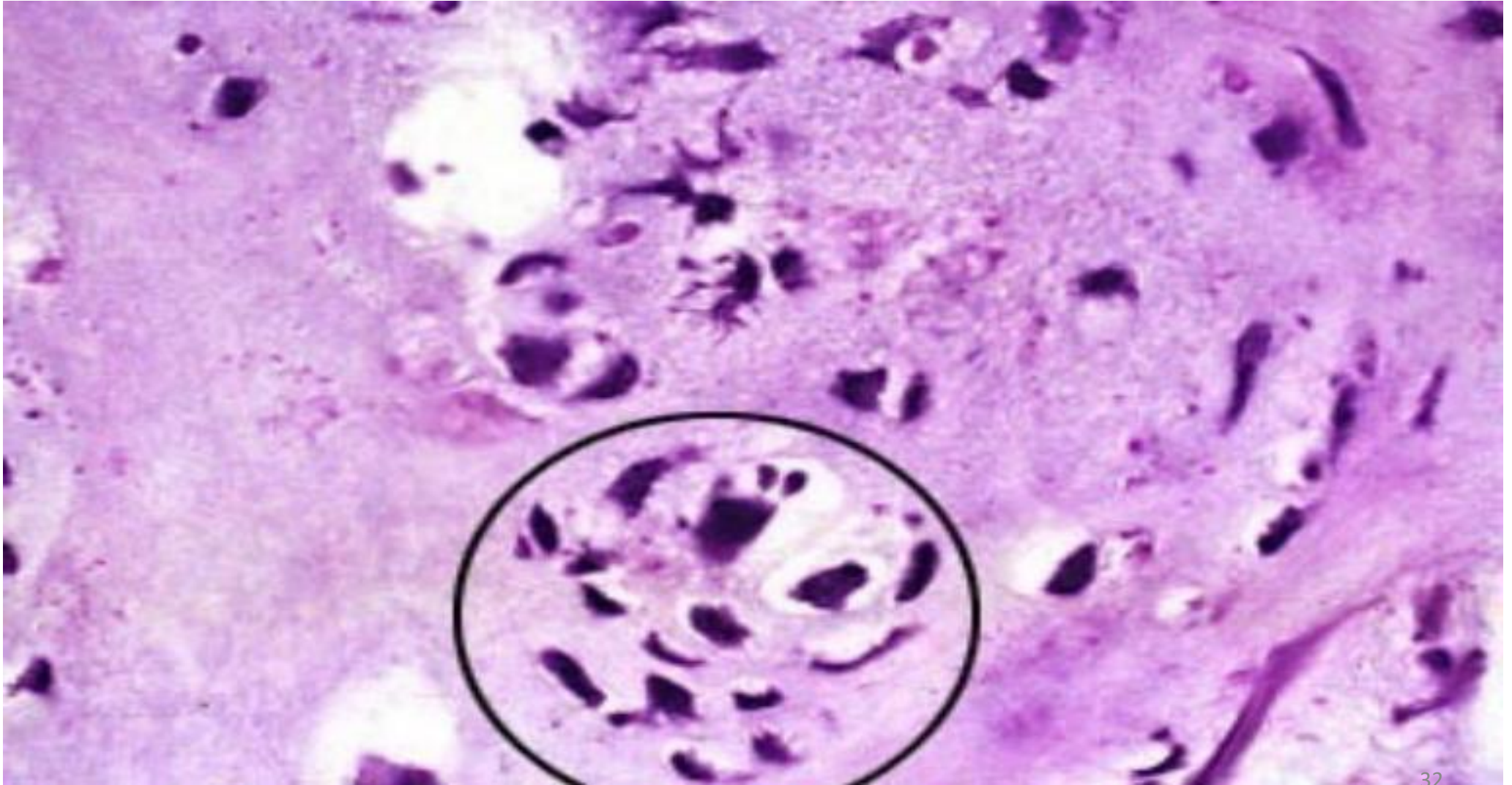
- In contrast to other tissues, hyaline cartilage is more susceptible to degenerative aging processes.
- Calcification of the matrix, preceded by an increase in the size and volume of the chondrocytes and followed by their death, is a common process in some cartilage.

2. **"Asbestiform" degeneration:** frequent in aged cartilage, is due to the formation of localized aggregates of thick, abnormal collagen fibrils.

3. **Cartilage Tumors:** cartilage cells can give rise to benign (chondroma) or malignant (chondrosarcoma) tumors.

- A notable histological finding is that multiple cells can be found within a single lacuna. In hyperplastic or normal cartilage, chondrocytes do not share their lacunae.
- When chondrosarcomas metastasize, they preferentially spread to the lungs.

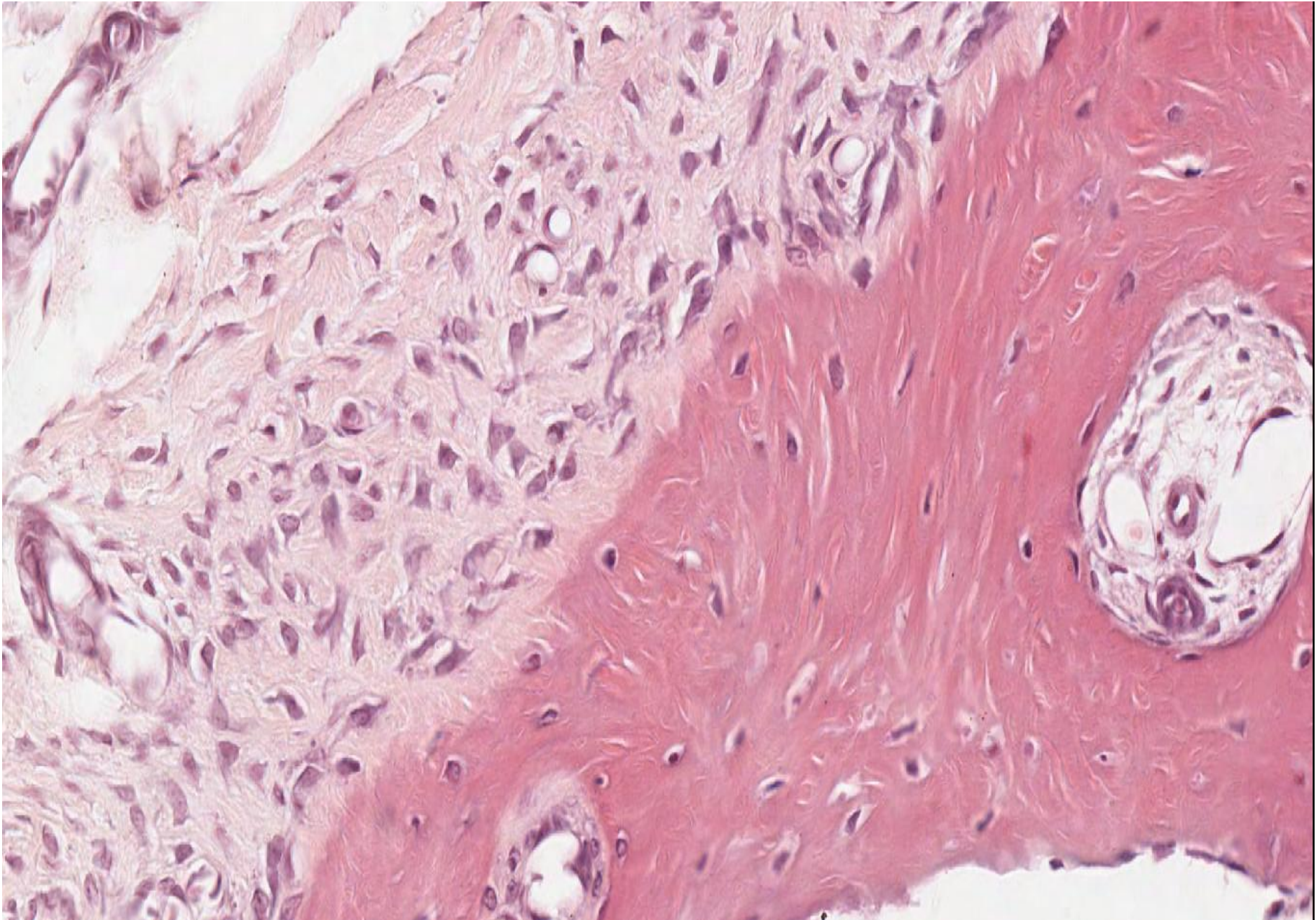
Bone, chondrosarcoma - High power: In this image, note the hyperchromatic and poorly differentiated neoplastic mesenchymal cells embedded in a cartilaginous matrix. In more differentiated tumors, distinct lacunae can be identified.



BONES:

- Bones are living organs composed of several types of connective tissues including the:
 1. **Periosteum** (double-layered CT membrane covering the external surface of the entire bone except for the joint surfaces)
 2. **Endosteum** (delicate loose areolar connective tissue membrane lining medullary & other bone marrow cavities)
 3. **Bone or osseous tissue**
 4. **Bone marrow tissues:**
 - i. **Reticular tissue (Red bone marrow)**
 - ii. **White Adipose tissue (Yellow bone marrow)**
- Based on the organ density, bone is classified as **Compact (dense) bone** and **Spongy (cancellous) bone**.

Section of bone showing Periosteum, osseous tissue and endosteum



Types of bones

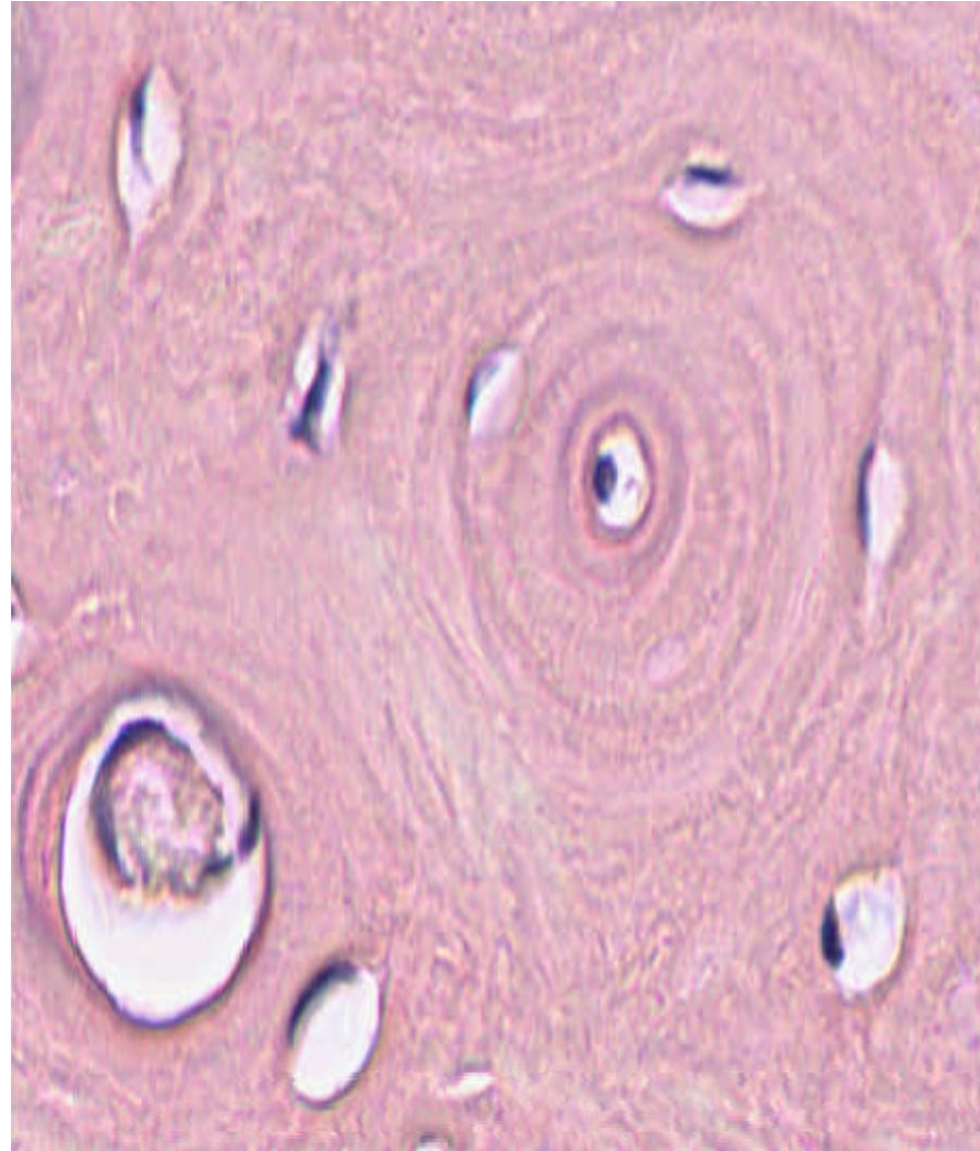
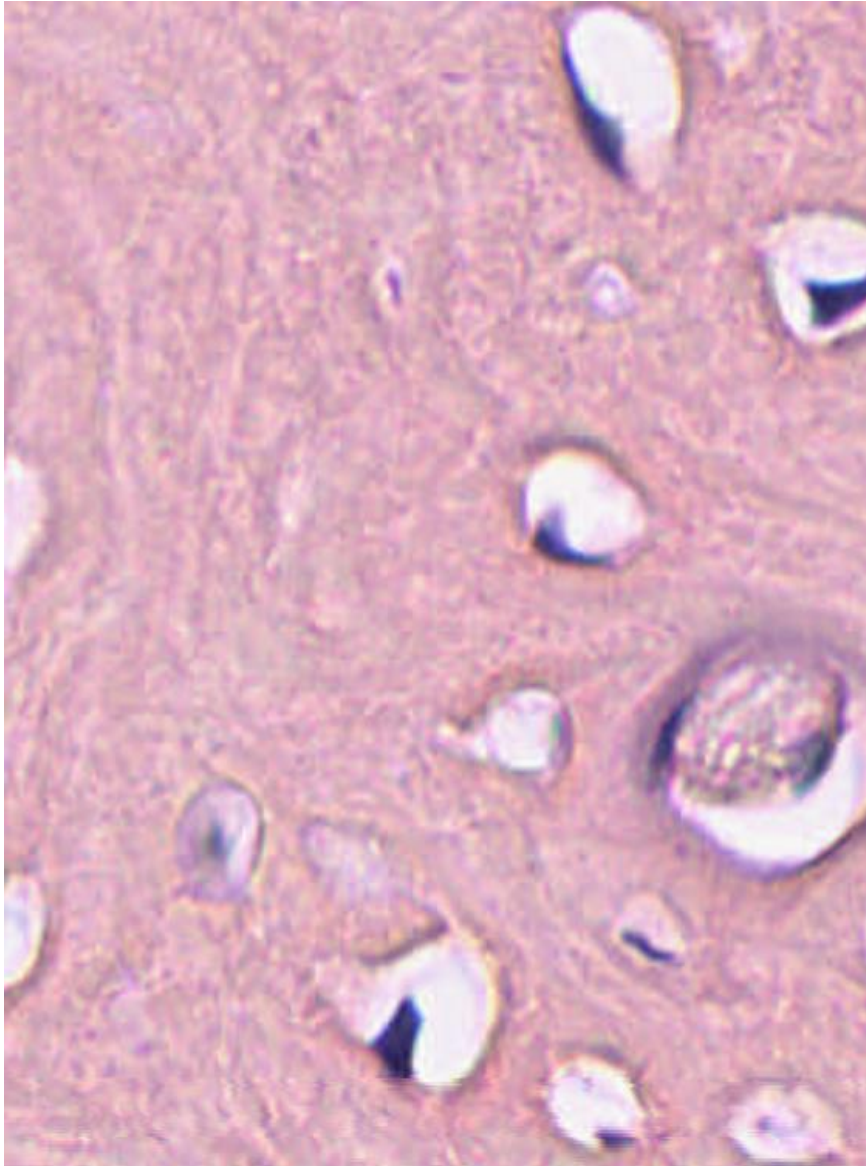
1. **Cancellous bone:** trabecular or spongy bone
2. **Compact bone:** dense bone

BONE TISSUE:

– Based on bone developmental maturity and organization of bone matrix & cells, bone tissue is classified as:

1. **Woven bone,**
 - named due to interlacing arrangement of its collagen fibers and cells
 - also called bundle bone or non-lamellar bone
2. **Lamellar bone:** collagen fibers and cells orderly arranged in lamellae.

Woven bone and Lamellar bone



BONE HISTOLOGY

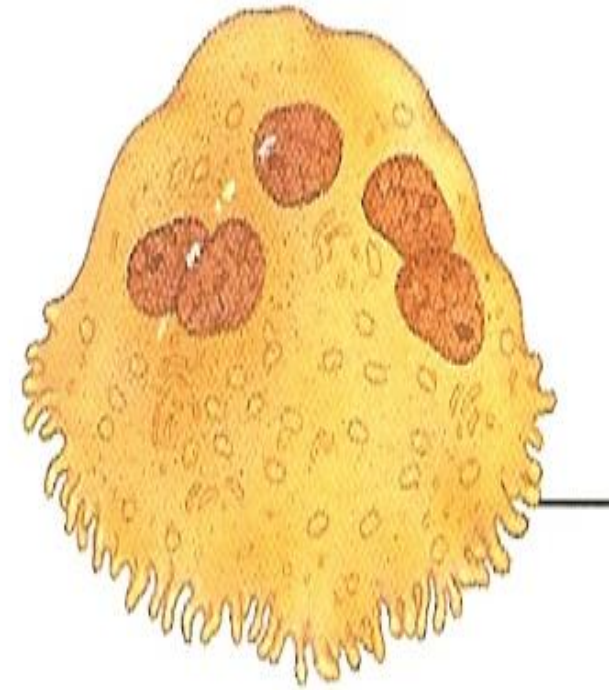
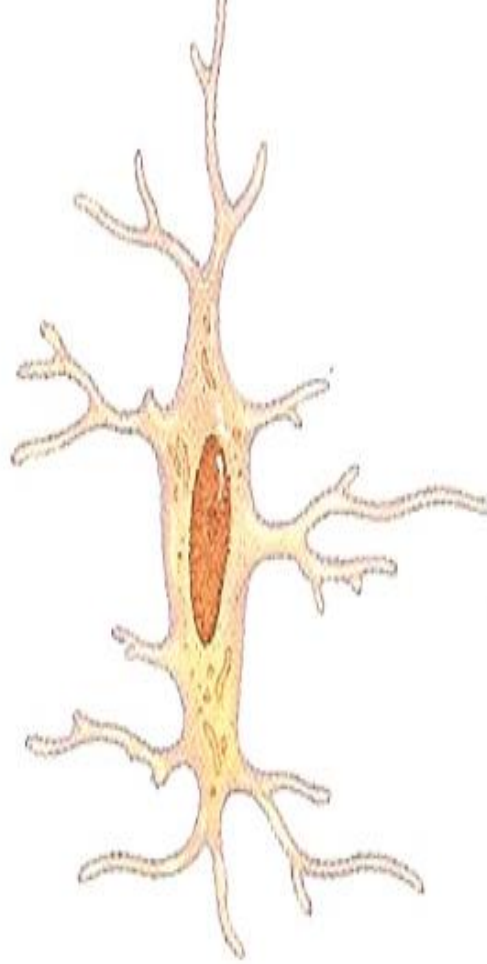
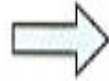
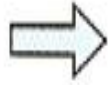
BONE OR OSSEOUS TISSUE

- Bone tissue is a type of connective tissue, hence consists of cells (bone cells) and a significant amount of extracellular matrix (bone matrix).
 - a. **Bone cells:** Osteoprogenitor cells (visible in developing bones only), osteoblasts, osteocytes, and osteoclasts
 - b. **Bone matrix** (with both organic and inorganic components)

Bone matrix.

Organic: collagen Type I fibers, proteoglycans and glycoproteins (e.g. osteocalcin and Osteonectin)

Inorganic: Ca hydroxyapatite $[\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2]$ crystals



Osteogenic cell
(develops into an
osteoblast)

Osteoblast
(forms bone
matrix)

Osteocyte
(maintains
bone tissue)

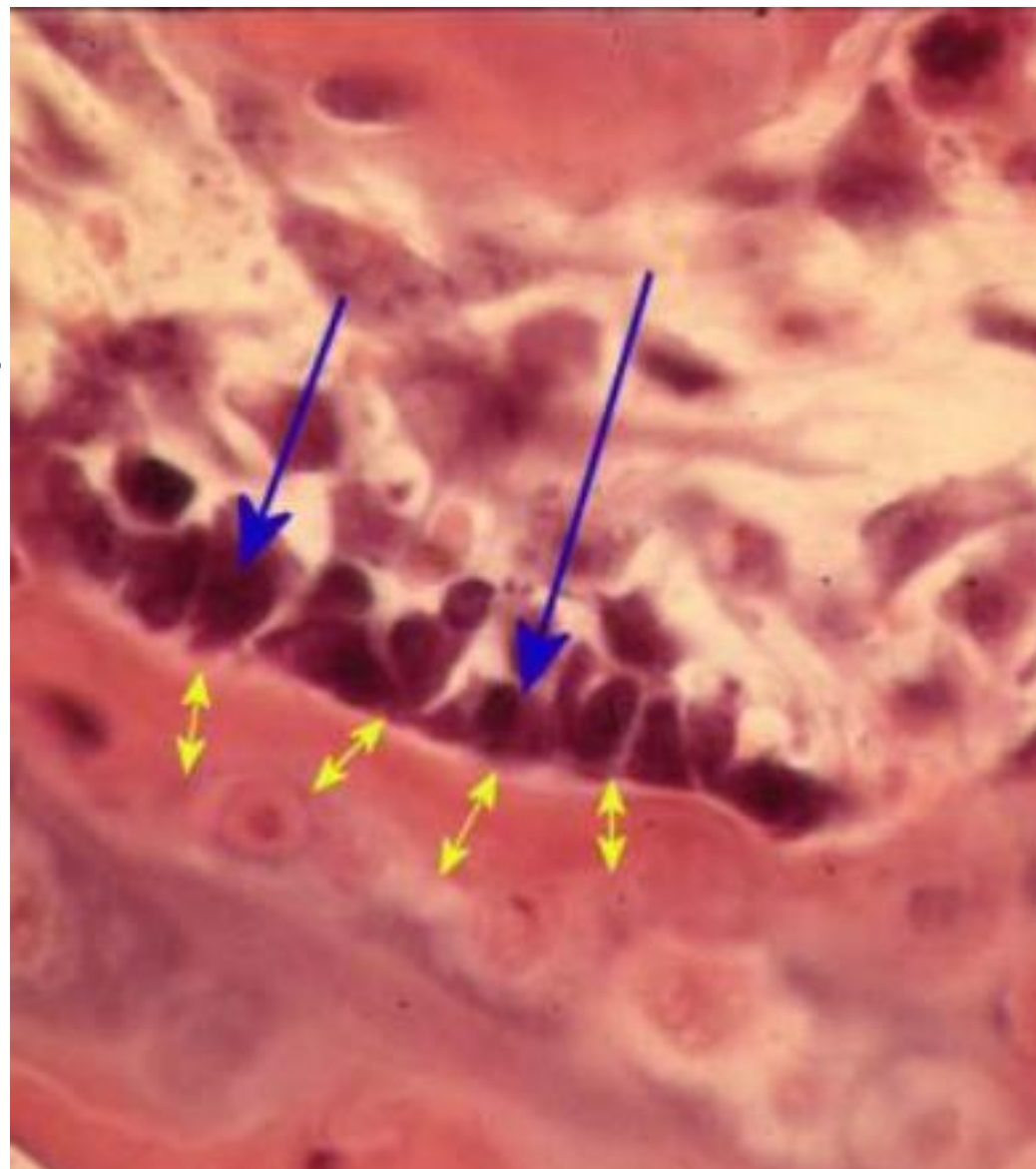
Osteoclast
(functions in resorption, the
breakdown of bone matrix)

1. Osteoprogenitor/Osteogenic Cell

- is derived from mesenchymal stem cells in the bone marrow that have the potential to differentiate into many different cell types including fibroblasts, osteoblasts, adipocytes, chondroblasts/cytes, and muscle cells.
- **Osteogenesis**, the process of new bone formation, requires a population of renewable osteoprogenitor cells (osteoblast precursor cells) that are responsive to molecular stimuli that transform them into bone-forming cells.
- The key factor that triggers differentiation of osteoprogenitor cells is a transcription factor called **core binding factor alpha-1 (CBFA1)**, which prompts the expression of genes that are characteristic of the phenotype of the osteoblast.
- Bone morphogenic proteins also play a role in the differentiation of osteoblasts.

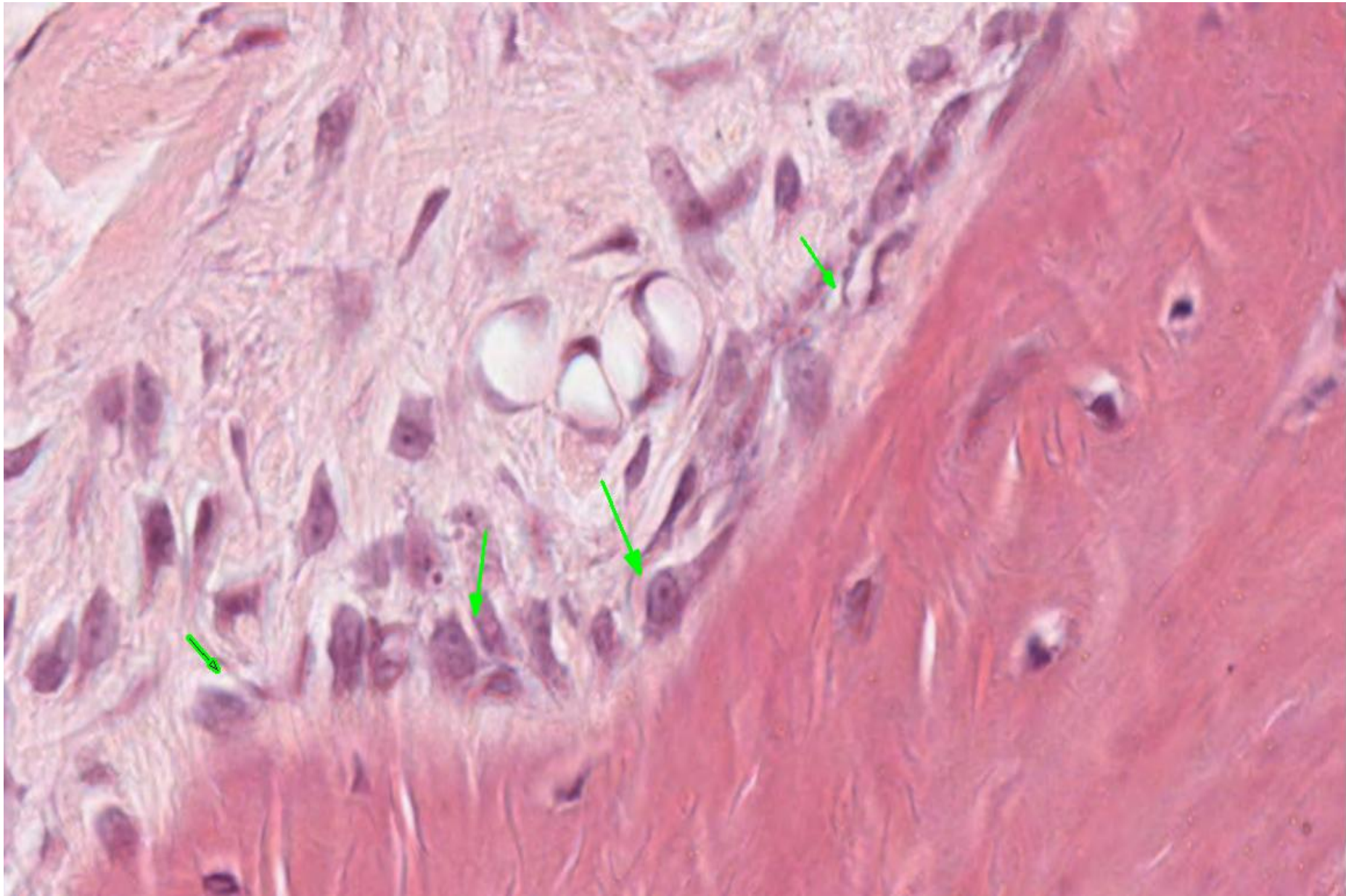
2. Osteoblasts

- Bone-building cells.
- Synthesize and secrete collagen fibers and other organic components of bone matrix.
- Initiate the process of calcification.
- Found in both the periosteum and the endosteum



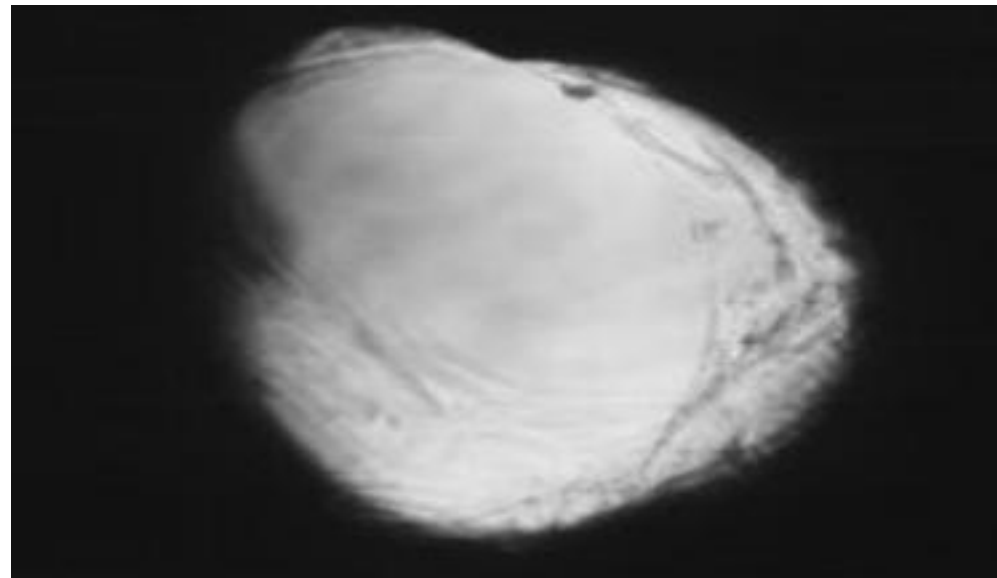
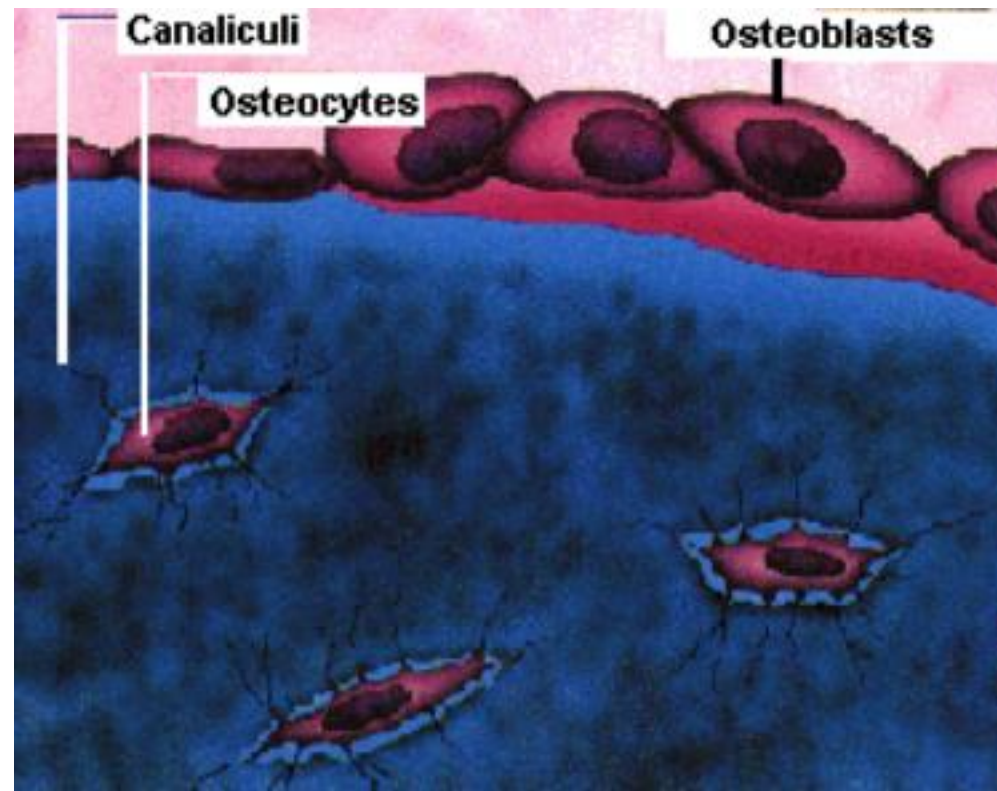
The blue arrows indicate the osteoblasts. The yellow arrows indicate new bone matrix (osteoid) secreted by the osteoblasts.

Osteoblasts

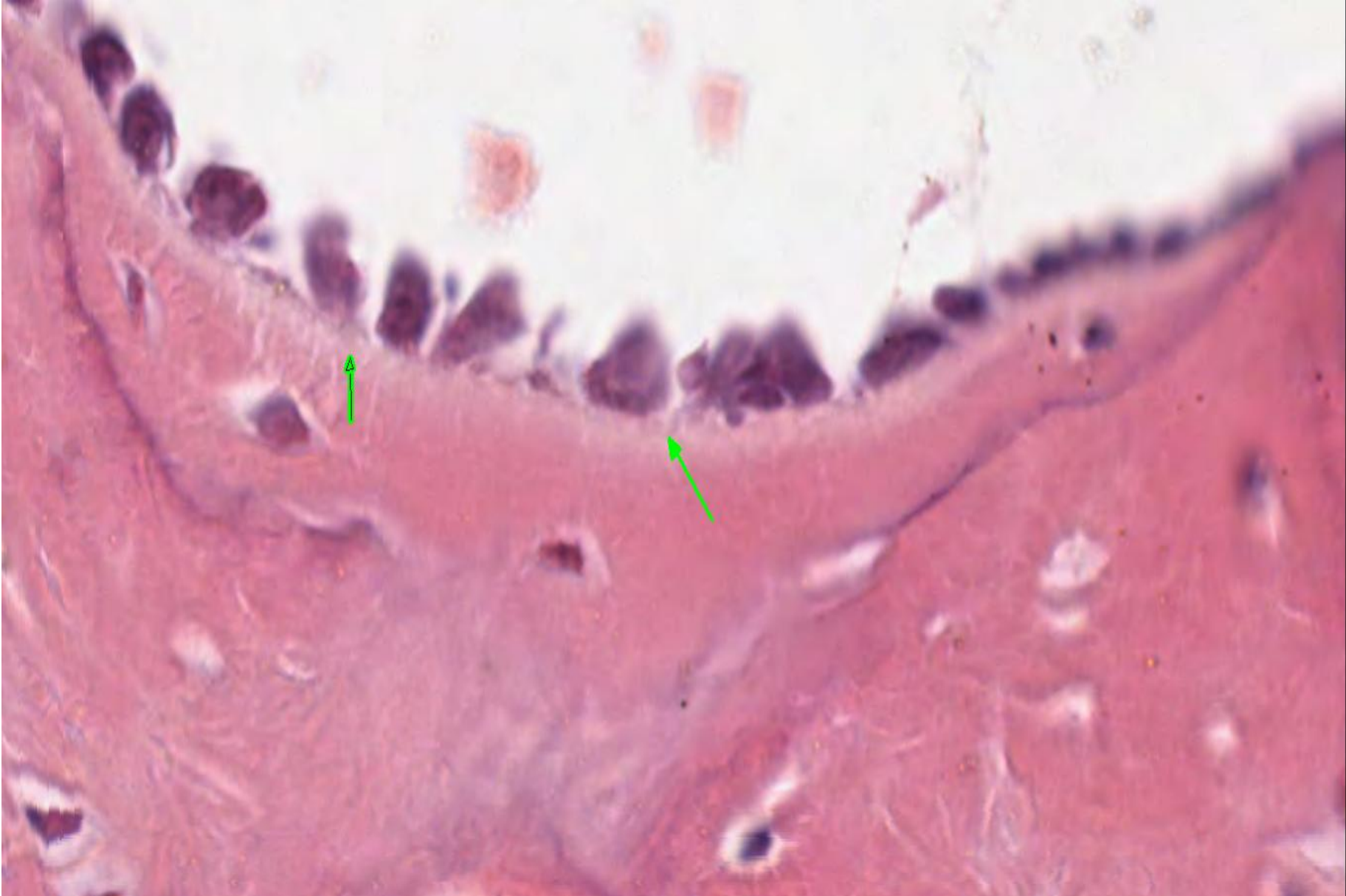


Osteoblasts

- Synthesize organic components of matrix (collagen type I, proteoglycans, & glycoproteins)
- The newly secreted organic matrix is called **osteoid**.
- Collagen forms strands of spiral fibers around which the inorganic matrix is deposited.
- Influence deposit of Ca^{++} , PO_4 .



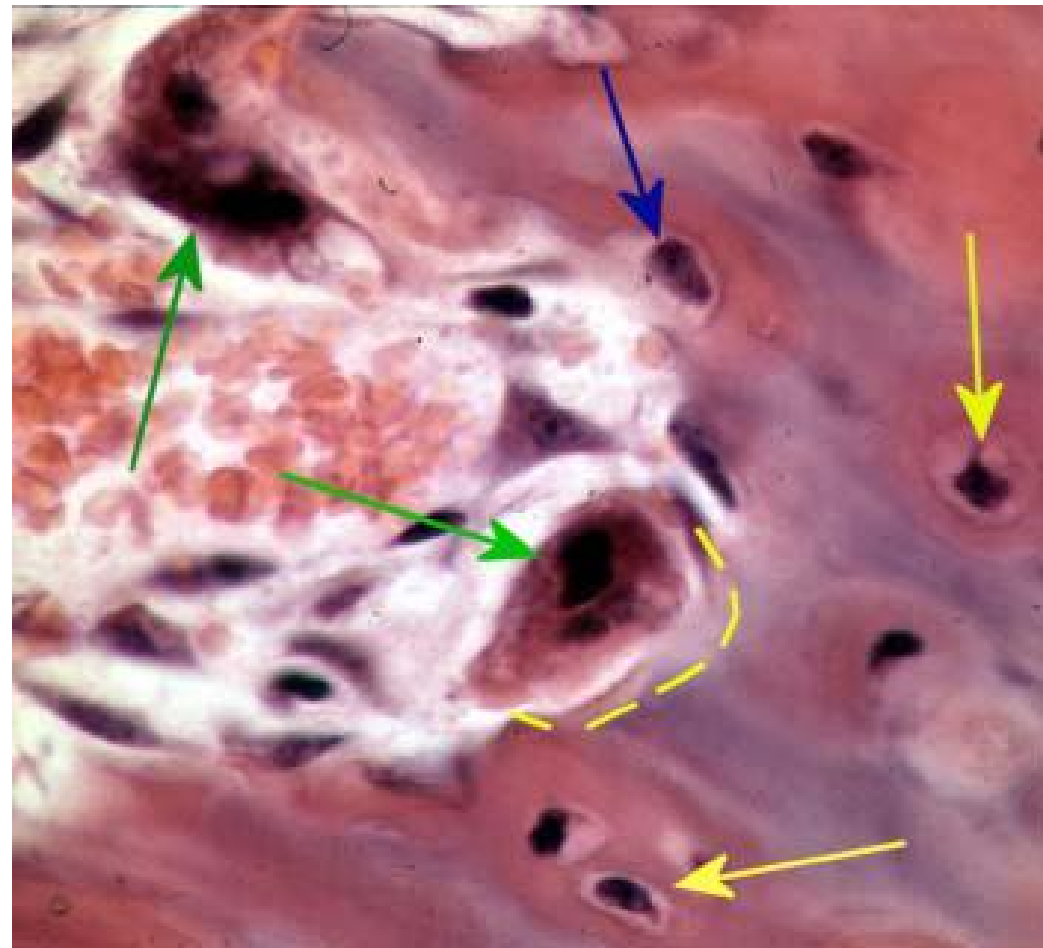
OSTEOID (arrows)



3. Osteocytes

- Mature bone cells.
- Osteoblasts that have become trapped within the secreted bone matrix.
- Responsible for maintaining the bone tissue by renewing bone matrix.

- On the right, notice how the osteocyte is “trapped” within the pink matrix

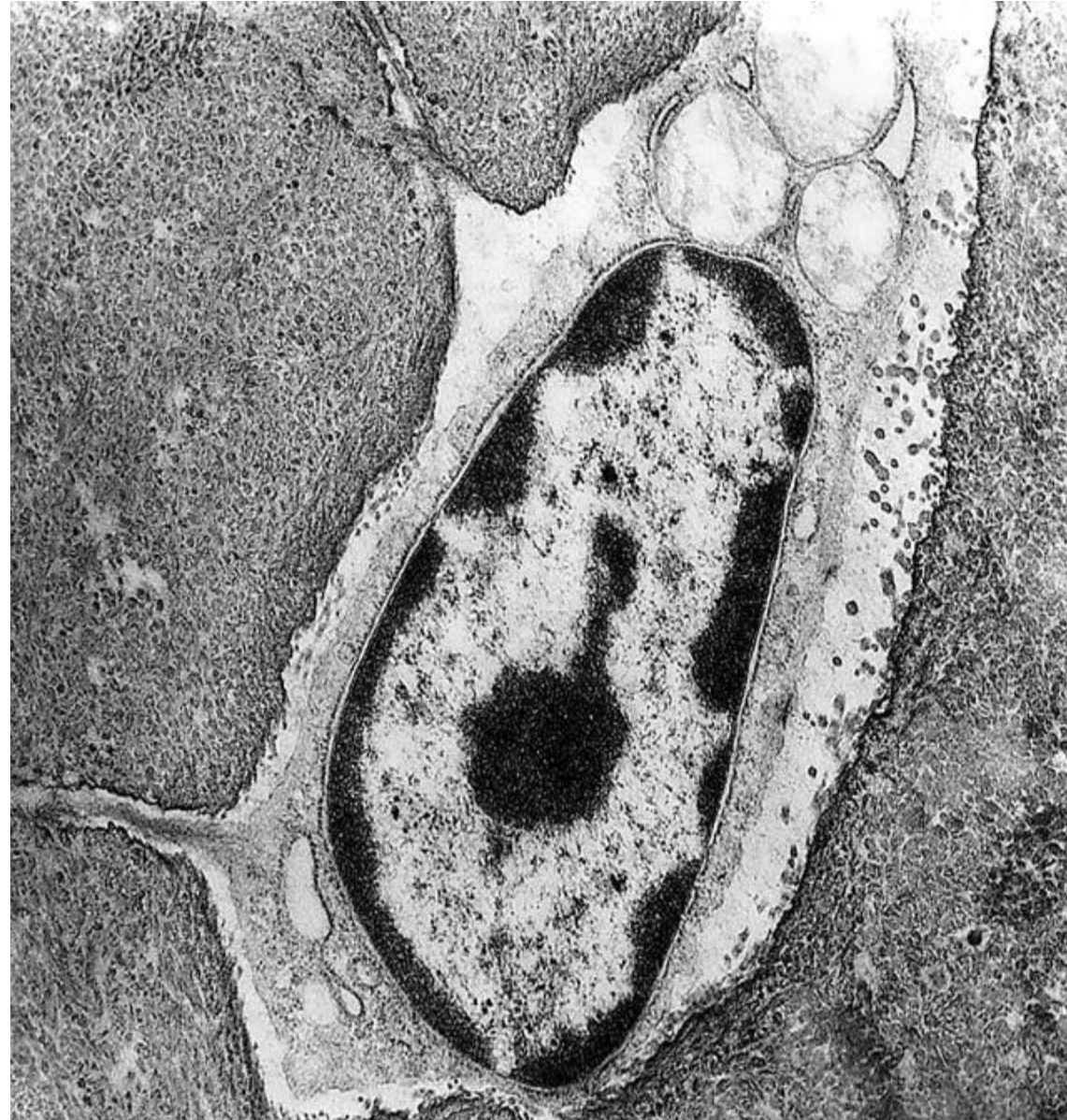


- Yellow arrows indicate **osteocytes** – notice how they are surrounded by the pinkish bone matrix (lacunae).
- **Blue arrow** shows an **osteoblast** in the process of becoming an osteocyte.

Osteocytes

- Mature bone cells that sit in lacunae
- Gap junctions between osteocytes provide nutrition (15 cells in a row)
- Maintain bony matrix; long lived cells
- Stimulated by calcitonin; inhibited by PTH

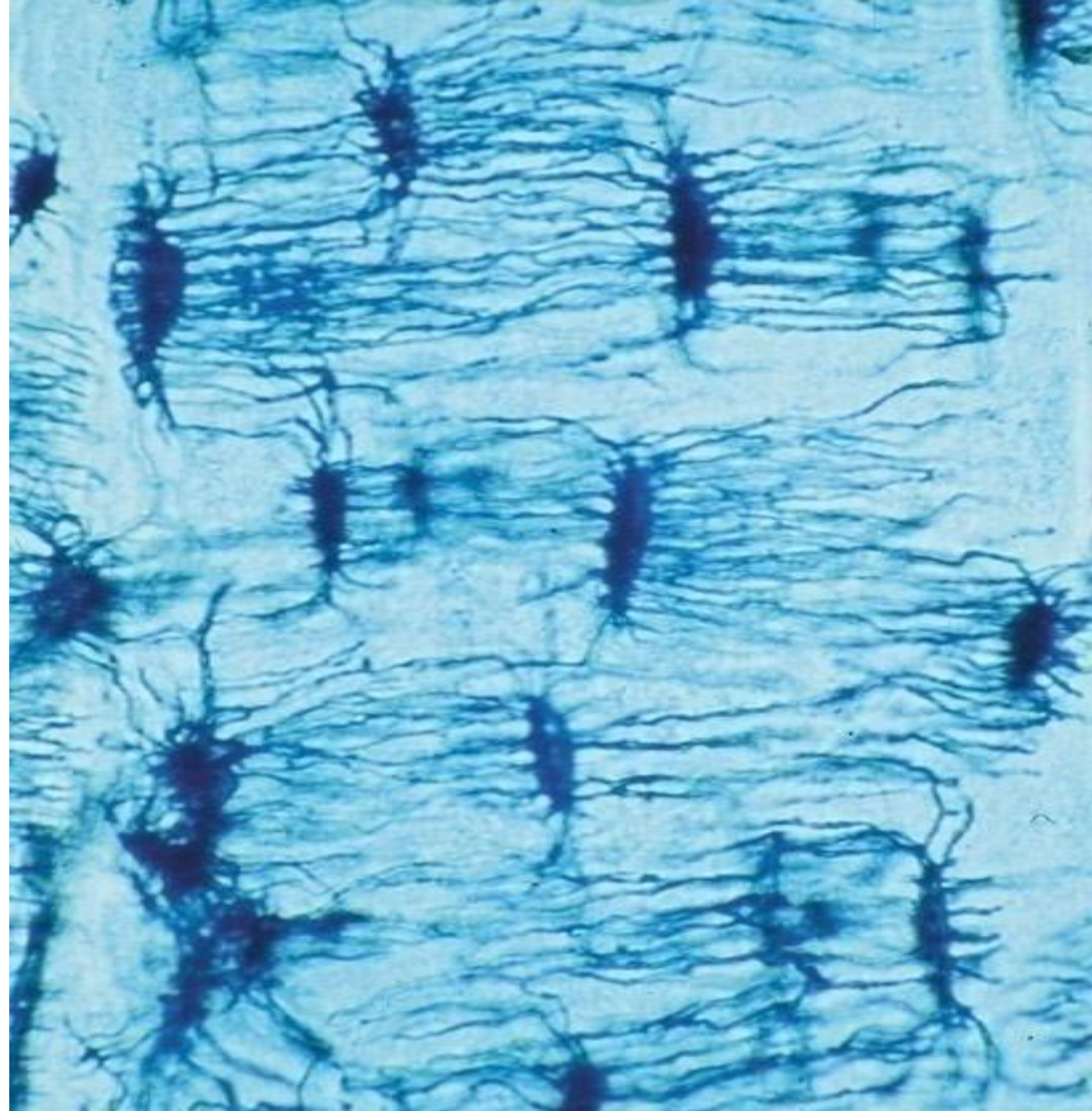
Osteocyte with Cytoplasmic Extensions



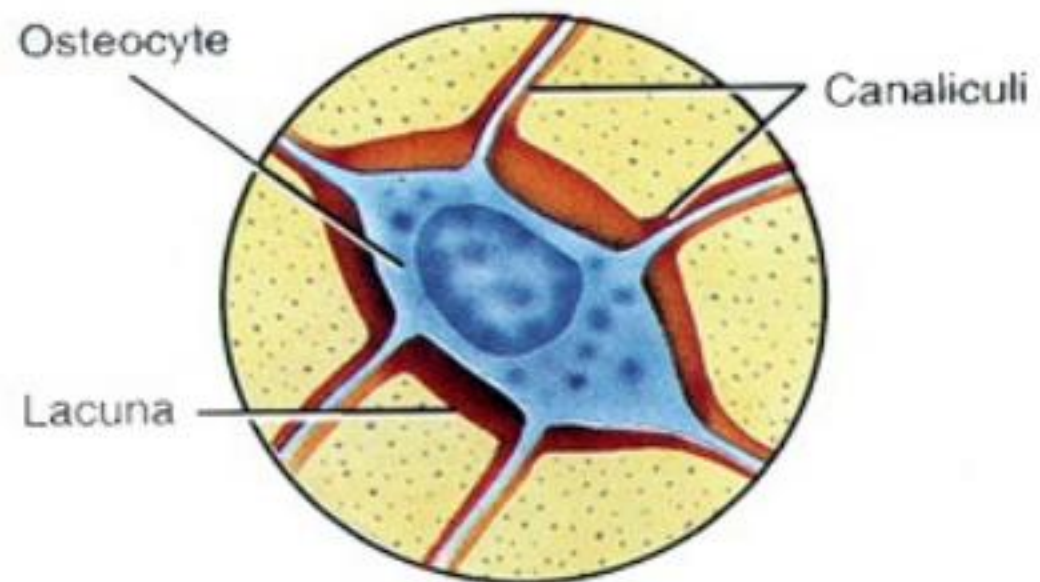
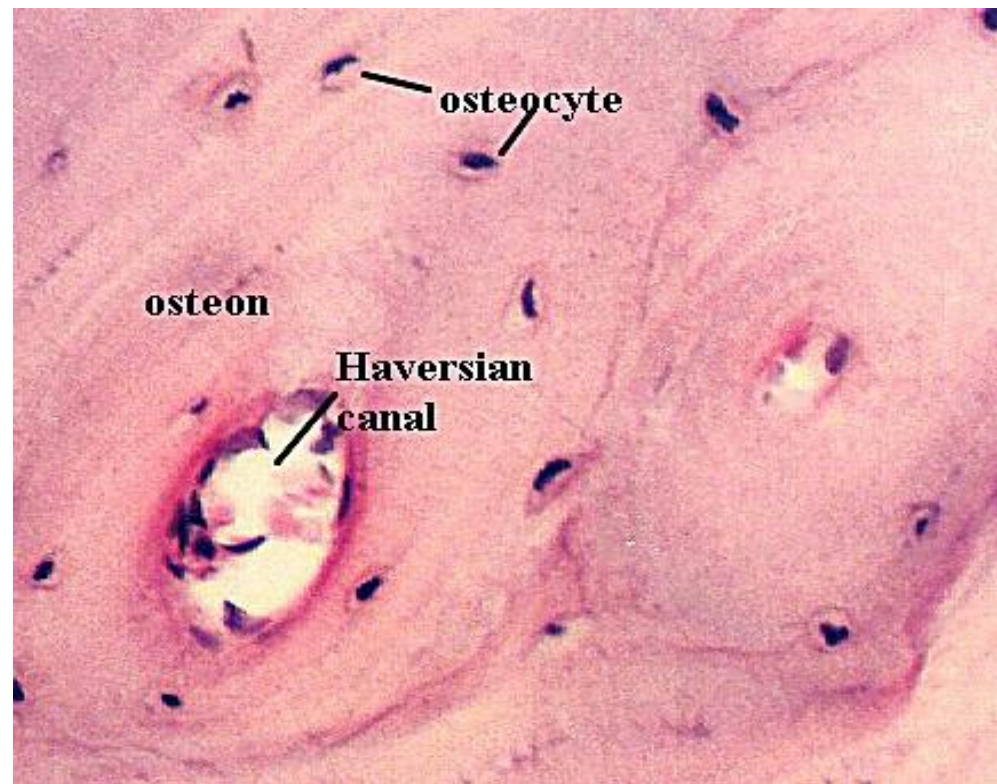
Electromicrograph of an osteocyte

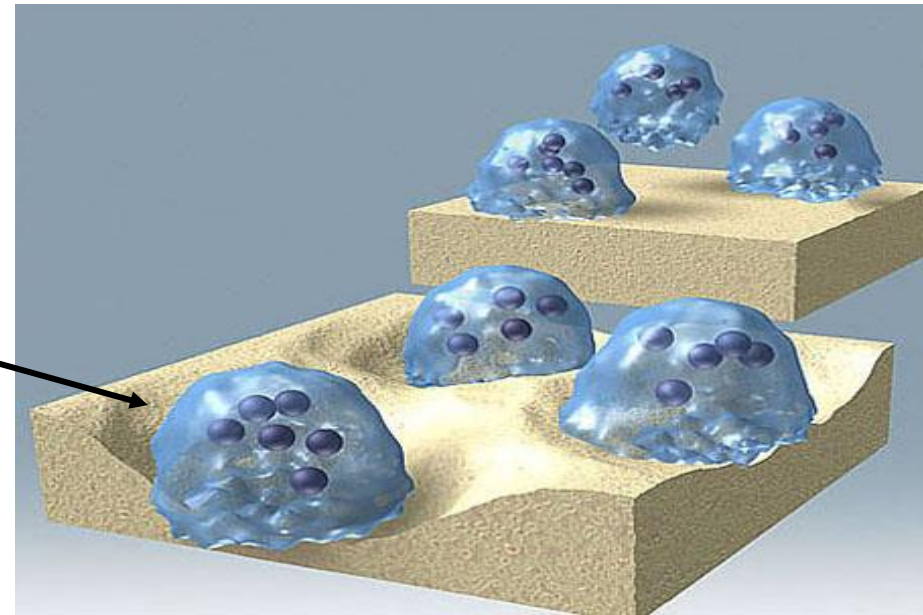
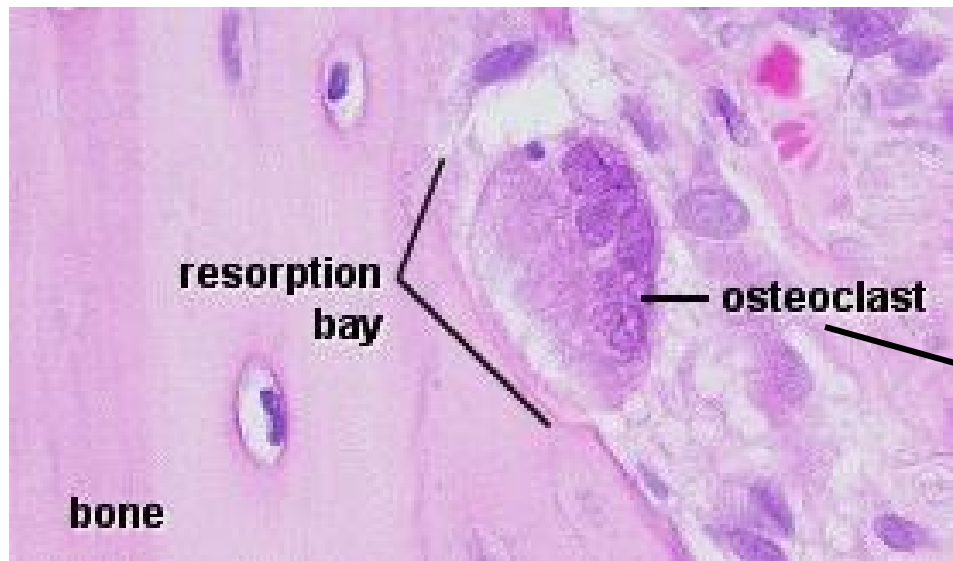
Lacunae and Canaliculi

- Photomicrograph of dried bone ground section, very thin.
- The lacunae and canaliculi filled with air deflect the light and appear dark, showing the communication between these structures through which nutrients derived from blood vessels flow.



- Spider-shaped osteocytes occupy small cavities known as **lacunae** at the junctions of the lamellae.
- Hair like canals called **canaliculi** connect the lacunae to each other and to the central canal.
- Canaliculi allow the osteocytes to exchange nutrients, wastes, and chemical signals to each other via intercellular connections known as **gap junctions**.



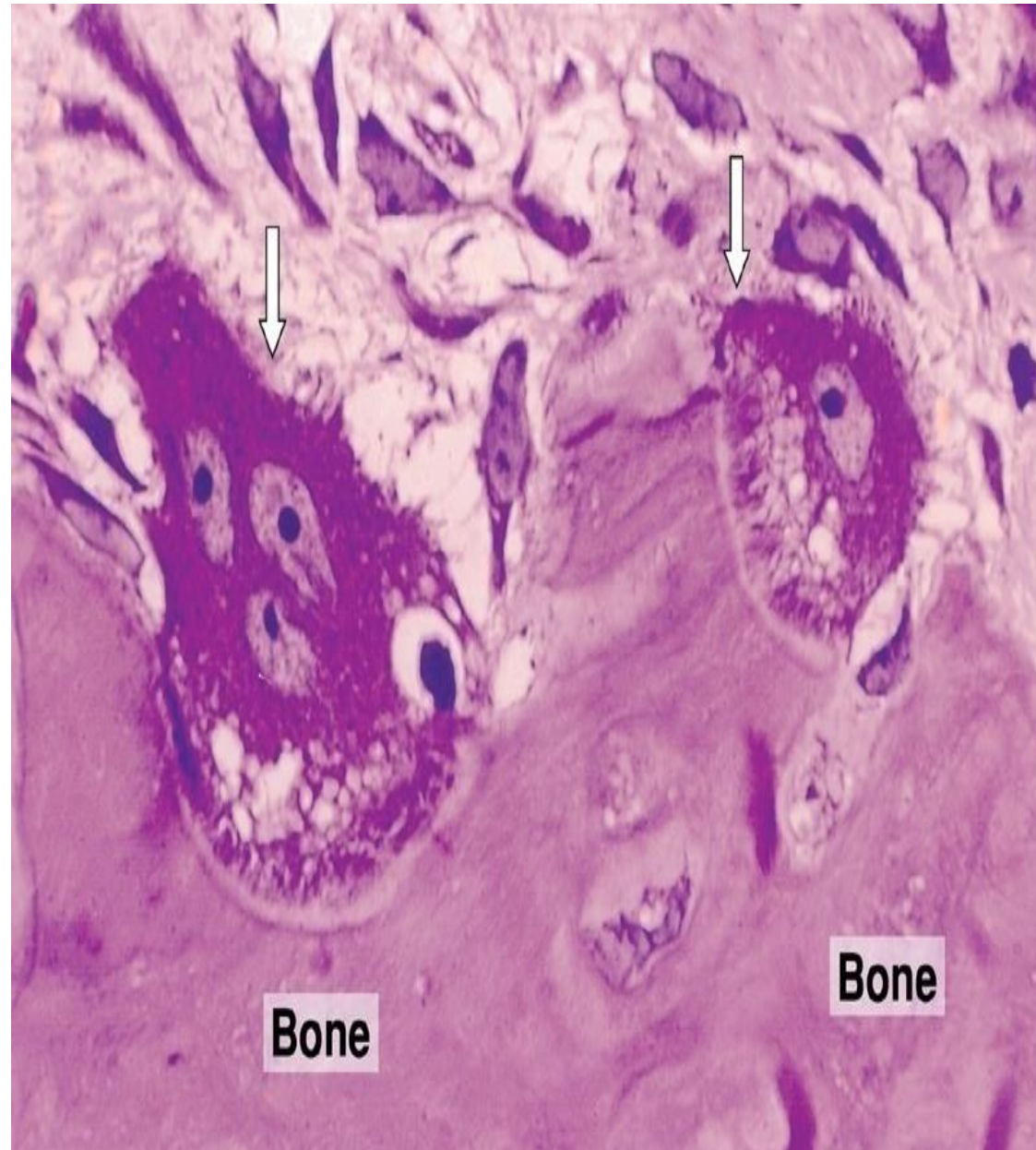


4. Osteoclasts

- Huge cells derived from the fusion of as many as 50 monocytes (or their precursors in the bone marrow).
- Cells that digest bone matrix – this process is called **bone resorption** and is part of normal bone growth, development, maintenance, repair or remodeling.
- Concentrated in the endosteum.
- On the side of the cell that faces the bone surface, the PM is deeply folded into a **ruffled border**.
- Here, the osteoclast secretes digestive enzymes to digest the bone matrix.
- It also pumps out hydrogen ions to create an acidic microenvironment that eats away at the matrix. *What advantage might a ruffled border confer?*
- Why do we want a cell that eats away at bone? (Hint: bone is a very dynamic tissue.)

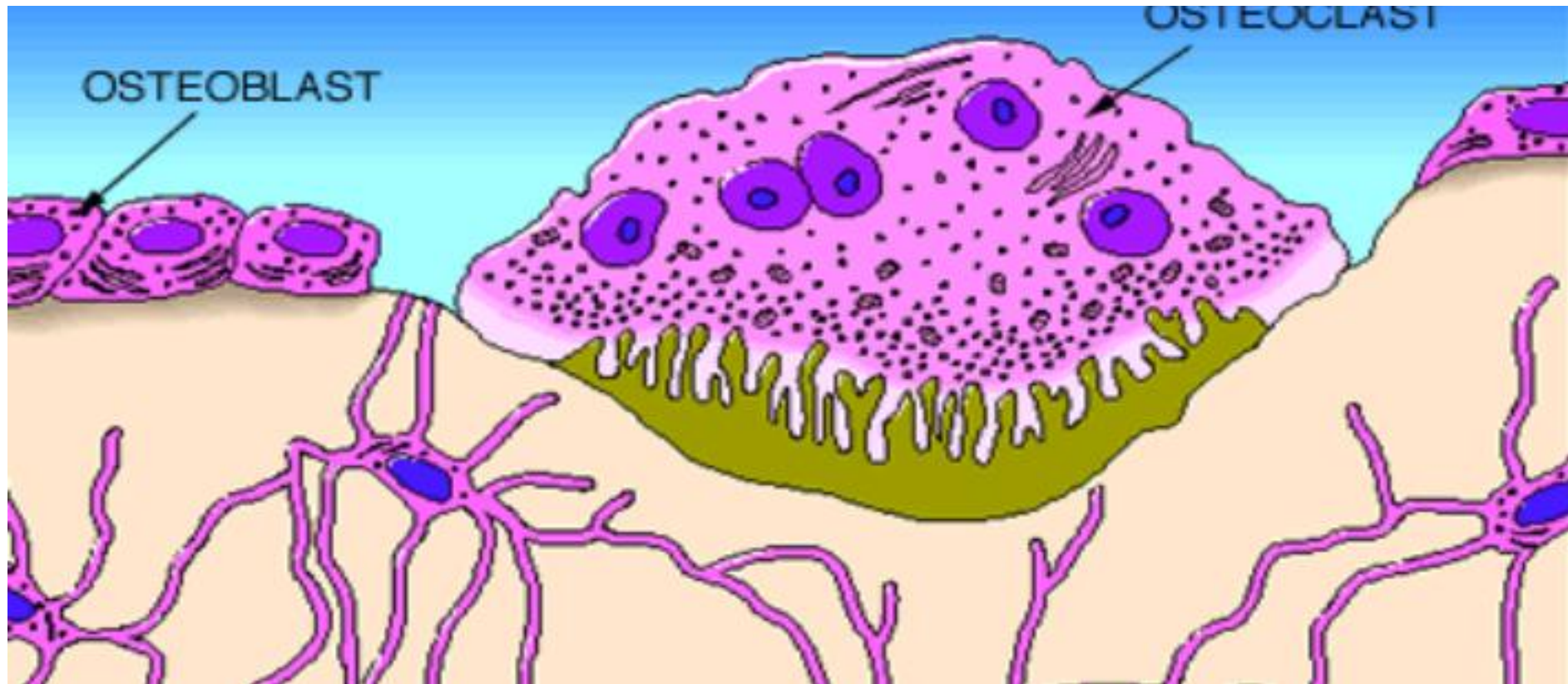
Osteoclasts

- Large, branched, motile cells
- Derived from monocytes; engulf bony material
- Sit in a depression of bone matrix known as **Howship lacunae (Resorption Bay)**
- Secrete enzymes that digest bone matrix when stimulated by active osteoblasts.

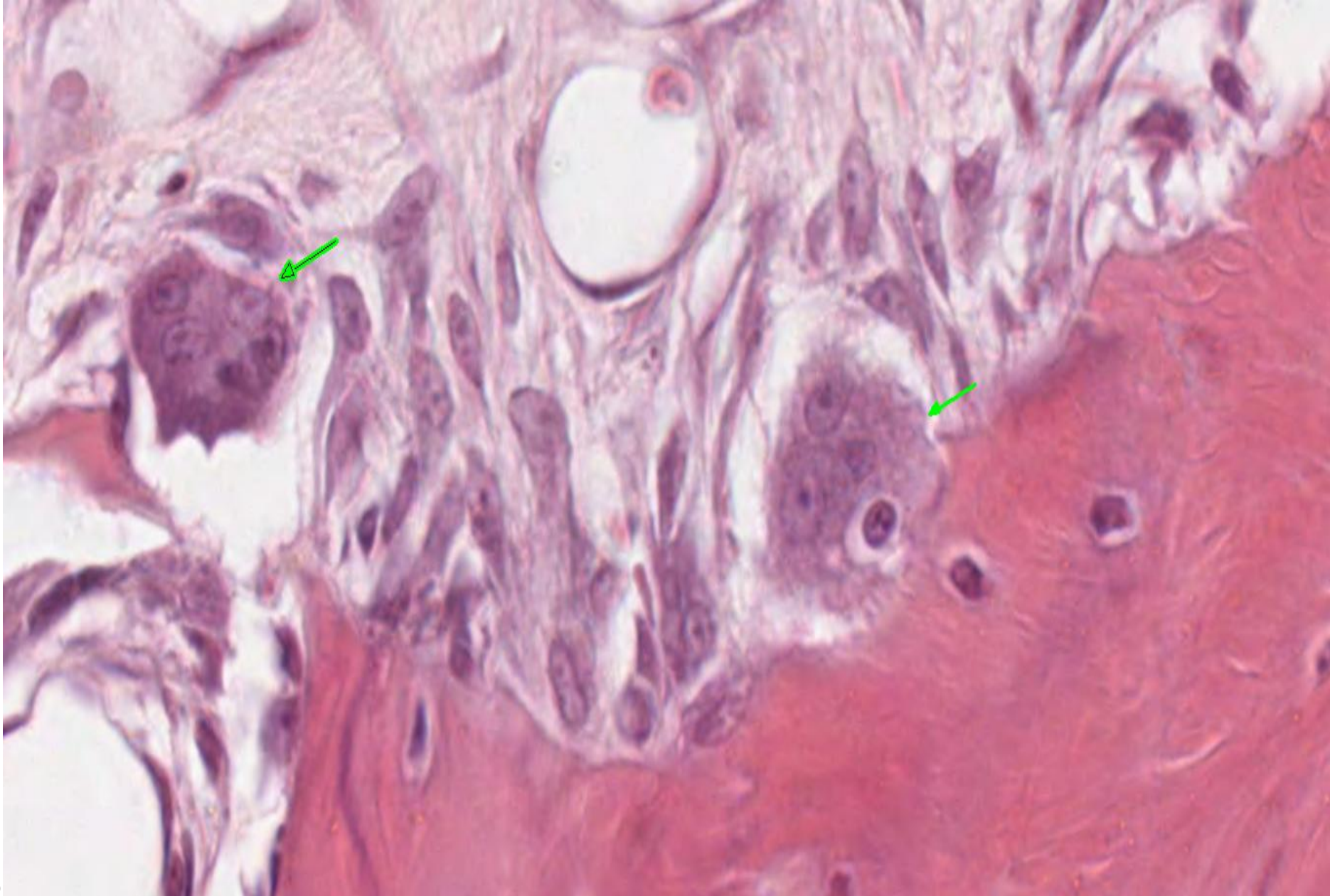


Here, we see a cartoon showing all the 3 cell types of mature bone cells: osteoblasts, osteoclasts and osteocytes.

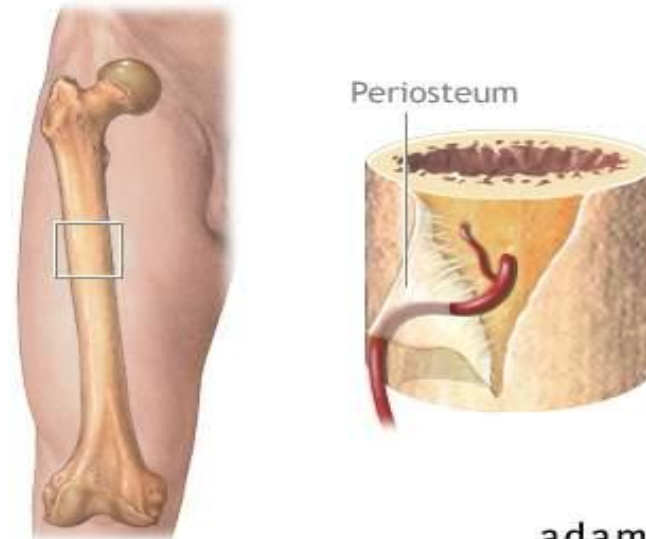
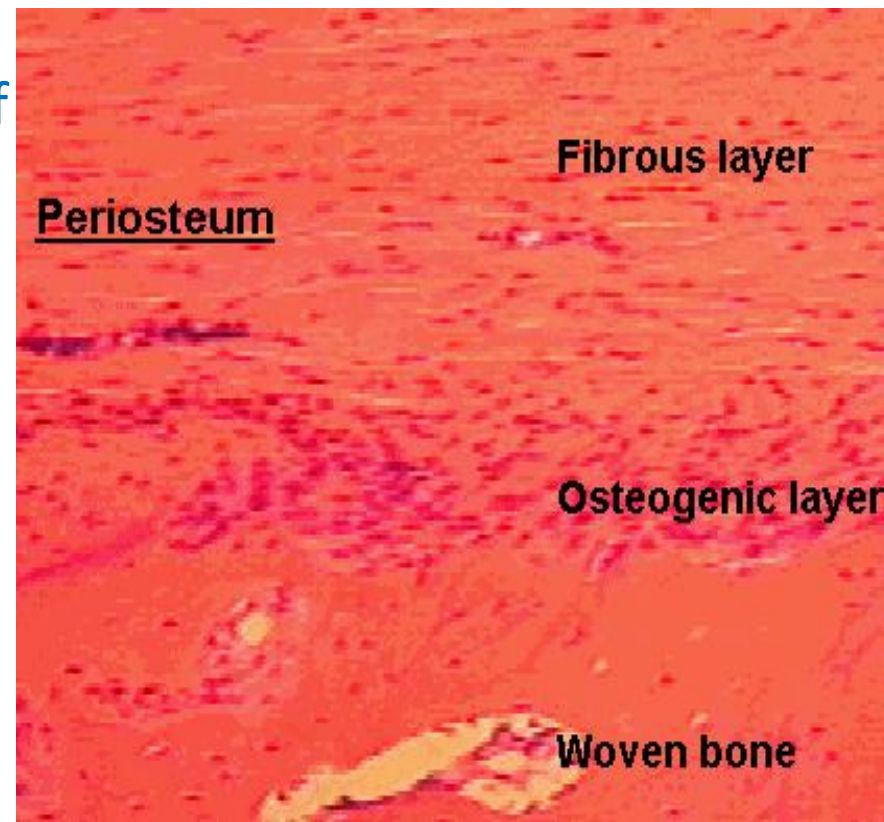
- Note the size of the osteoclast (compare it to the osteoblast), and note the ruffled border.
- Why is there a depression underneath the osteoclast?
- What is the name of the third cell type shown here?
- What do you think the tan material represents?

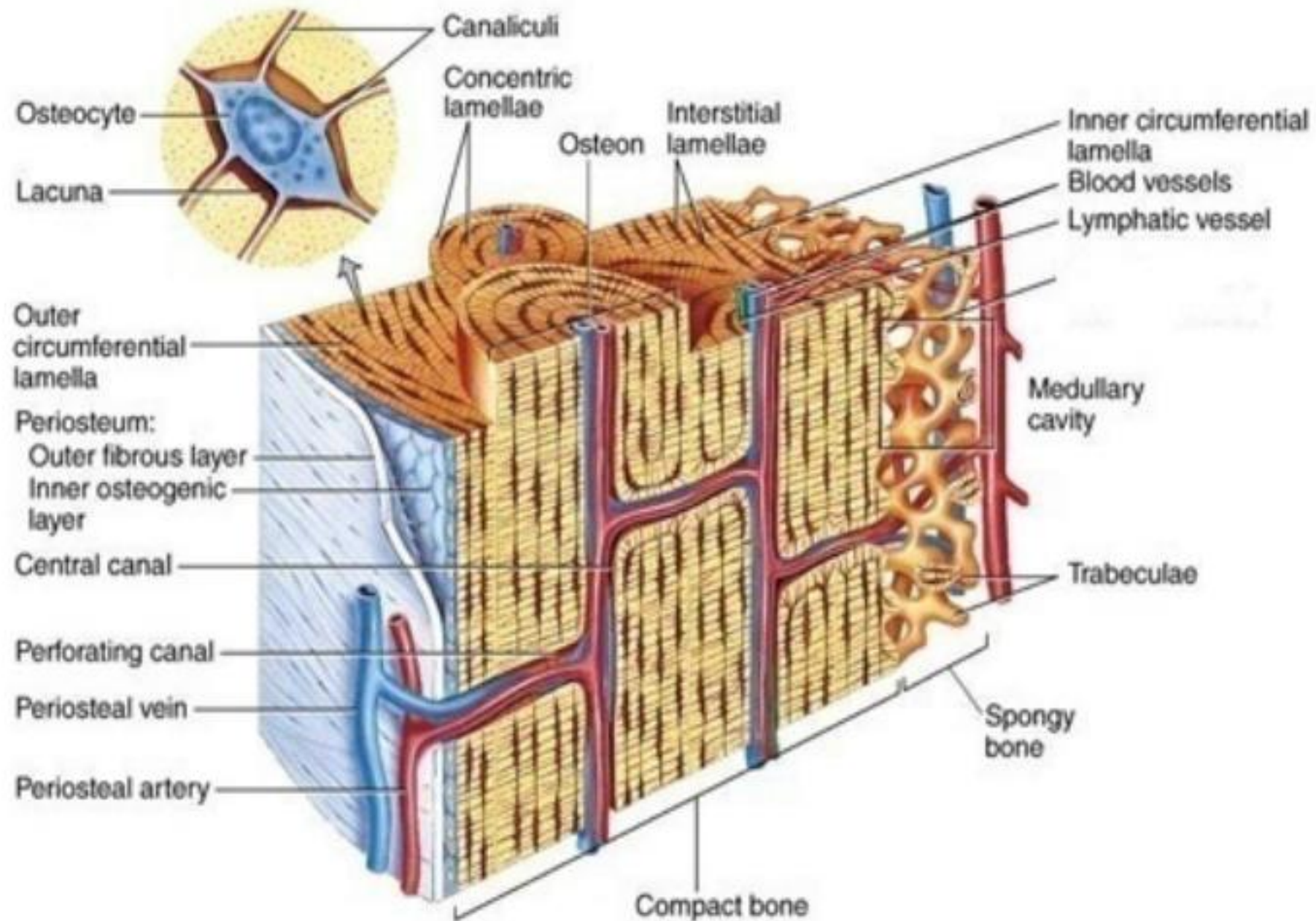


OSTEOCLASTS (arrows)



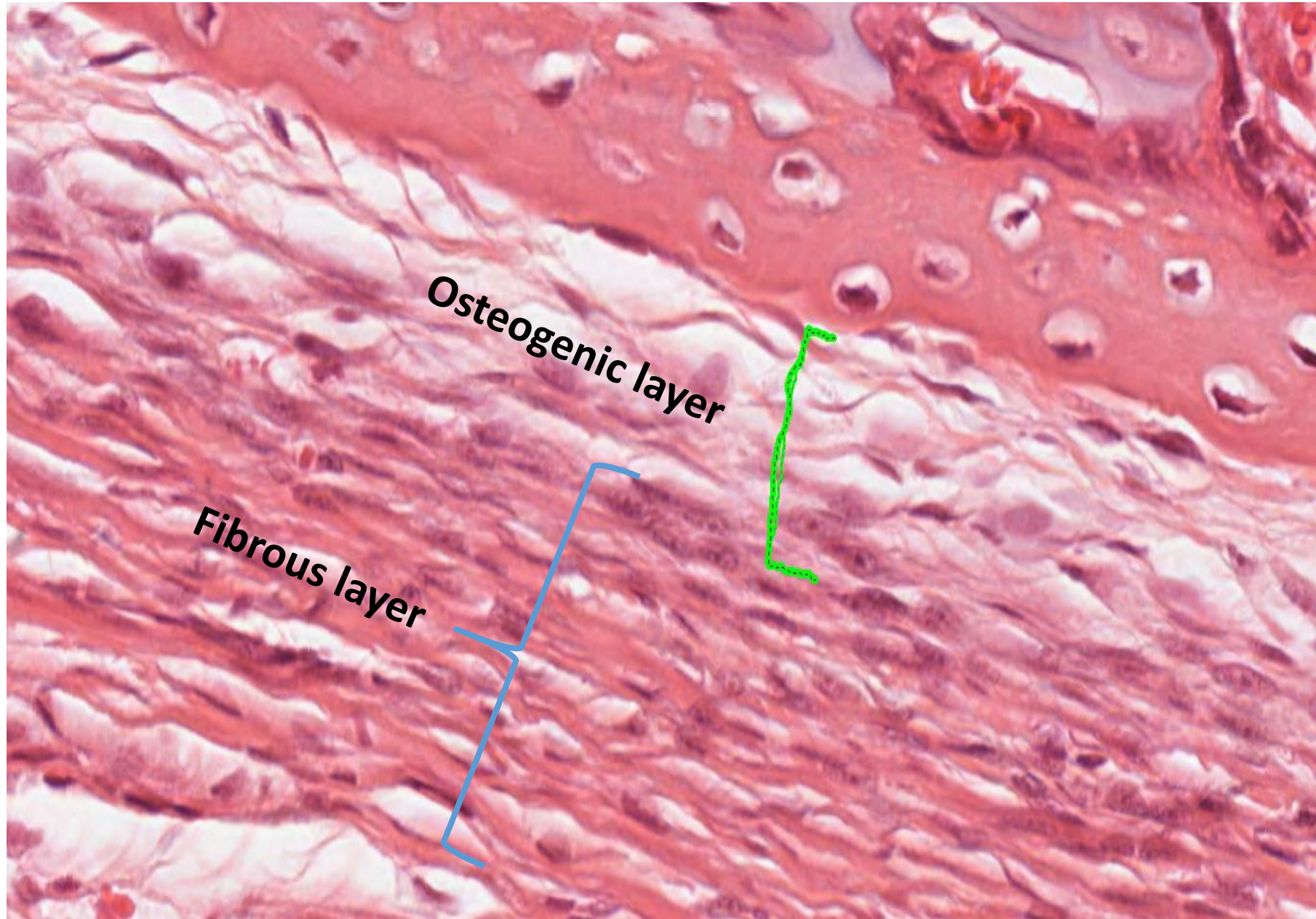
- The external surface of the entire bone except for the joint surfaces of the epiphyses is covered by a double-layered membrane known as the **periosteum** with an:
 1. Outer **fibrous layer** of dense collagenous connective tissue.
 2. Inner cellular, **osteogenic layer** containing **osteoprogenitor cells** and **osteoblasts**.
- Periosteum is richly supplied with nerve fibers, lymphatic vessels and blood vessels.
- These enter the bone of the shaft via a **nutrient foramen**.
- Periosteum is connected to the bone matrix via strong strands of collagens called **Sharpey's fibers**.





(a) Osteons (Haversian systems) in compact bone and trabeculae in spongy bone

IMMATURE PERIOSTEUM



- Internal bone surfaces and marrow cavities are lined by a delicate connective tissue membrane known as the **endosteum**:
- Covers the trabeculae of spongy bone in the marrow cavities and lines the canals that pass through compact bone.
- Contains both osteoblasts and osteoclasts.

